

The Architecture of the Collatz Space: Computational Mapping and Large-Deviation Theory

Sergey A. Zadorozhnii

August 2026

Abstract

Part I is a large-scale computational map of the Collatz space: we discover **Zone 2**, a family of 913 numbers (71–87 bits) whose trajectories merge into a single 75-bit node $x^* = 20152090995747160937051$ within ≤ 7 odd steps and then follow an identical path to a 140-bit peak; an isolated second path (Barina’s number); a continuous archipelago of confluence centers for every integer peak 14–51 with the scaling law $\text{bits}(c) = 0.498 \text{peak} + 6.29$; and a dead zone 88–170 bits free of ratio anomalies, confirmed by four independent searches. We further establish, by a Baker-type Diophantine argument combined with the Barina verification frontier 2^{68} , the *exclusion of all non-trivial cycles of length $d \leq 10^6$* ; a fixed-point census confirms 0 non-trivial solutions at every reachable length; and a 3-adic dimension-collapse argument shows that cyclic shift-words occupy a measure-zero set. Part II supplies the theory: a large-deviation duality between typical decay and rigid structures (F1/R1), soliton dominance (S1), the dressed-vacuum anatomy of centers (I1), the caustic log-midpoint law (C1), and the CRT obstruction to constructive synthesis (M1). We prove an exact 2-adic transport theorem, an unconditional Hausdorff dimension drop for the exceptional set at fixed barriers, and bridge to Shaik’s [8] unconditional natural-density polylogarithmic descent. Finally, we report the *Gate 2* campaign: a systematic falsification of every candidate pointwise transport mechanism (TV–Fourier restart, renewal closure, and Tree-Wasserstein multiblock contraction), confirming that the Archimedean–2-adic wall stands and the pointwise martingale/transport route is closed. Together the parts establish the paradigm: *crystals can be found, but not grown*. The pointwise Collatz conjecture is not proved; the five types of statement the word “Collatz” conflates are cleanly separated.

Contents

I	The Computational Map	1
1	Introduction and Chronology of the Study	2
1.1	The Collatz Problem	2
1.2	Starting Point: David Barina’s Results	2
1.3	Phase 1: Collatz Crystal Hunter and Markov Chains (February 2026) . .	2
1.4	Phase 2: Statistical Analysis of 30 Million Numbers (Early March 2026) .	3
1.5	Phase 3: From Statistics to Structure (March 2026)	3

2	Notation and Accelerated Dynamics	3
2.1	2-adic Interpretation	3
3	Base Layer: Family A	4
4	Zone 2: The Only Confirmed Major Confluence Class	4
4.1	Discovery	4
4.2	Table of Zone 2 Representatives	4
4.3	Two-Layer Invariants: Core and Adapter Shell	5
4.4	Confluence Point x^*	5
4.5	Reverse Tree of x^* and Full Size of Zone 2	6
4.6	Grammar of the Shift-Vector	7
4.7	The (1, 1, 2) Equilibrium and the Attractor $4/3$	7
5	Barina's Number: An Isolated Second Path to Peak 140	7
6	Number 27 as a Miniature Zone 2	7
7	Dead Zone 88–170 Bits	8
7.1	How the beam search probed the dead zone	8
7.2	Probabilistic Anatomy of the Dead Zone	9
7.3	The Algebraic Impossibility of Shadow Zones	9
7.4	The Control-Budget Obstruction to Constructive Macro-Solitons	10
8	Refutation of Zone 3	10
9	Archipelago of Confluence Centers	10
9.1	Systematic Census (Peaks 10–200)	10
9.2	Two Classes of Centers	12
9.3	Trends and Transitional Centers	12
9.4	Depth-Resolved Hit Rate: Class A as a Singular Class	12
9.5	Centers Do Not Cover the Space	12
10	Spectrum of Chords: From Number 27 to Zone 2	12
11	Summary Map of the Collatz Space	12
12	The Big Picture	13
13	Computational Complexity	14
14	Open Questions	14
15	Conclusion of Part I: Crystals Cannot Be Grown	14
II	Large-Deviation Theory	14
16	Introduction	14
17	Duality of the Typical and the Rigid	16

18 Forward and Backward LDP	16
18.1 Step A: Uniform spectral gap and large deviations of the tilted 3-adic flow	16
18.2 Step B: The Doob h -transform and shadowing LDP	18
19 Gate 2: Systematic Falsification of Pointwise Transport Mechanisms	23
19.1 G2.1: TV–Fourier restart (falsified)	23
19.2 G2.2: Renewal closure (falsified)	24
19.3 G2.3: Tree-Wasserstein multiblock contraction (falsified)	24
20 Cycle Exclusion	24
20.1 Fixed-point census (experiment 1.1)	24
20.2 High-mean admissibility: dimension collapse (experiment 1.2)	25
20.3 Baker-type exclusion up to length 10^6 (experiment 3.1)	25
21 Caustic Scaling	26
21.1 The Caustic Balance: Detailed Balance at the Waist (Model C1)	26
21.2 Phases 4–5: Solitons as Tilted-Typical Paths, Centers as Dressed Vacua .	27
21.3 Macro-obstruction to reverse synthesis	27
22 Front C and Fine-scale mixing	28
22.1 Two Fourier regimes and the worst-case barrier	30
22.2 Honest status of the programme	31
23 Family Dichotomy and Universal 2-adic Mechanics	32
23.1 The threshold $\delta(b) = \log 4 / \log b$	32
23.2 Numerical validation of the dichotomy	32
23.3 Septembrino’s observations as 2-adic arithmetic	33
23.4 Connection to Allikvere’s transport mechanism	33
23.5 Falsification of the Sum-Conditioned Restart Hypothesis	34
23.6 Exact Topological Transport in the 2-adic Metric	34
23.7 Martingale Validation of the Survival Model	35
24 Synthesis of the Two Heads: Exact 2-adic Transport and Sum-Conditioned 3-adic Mixing	35
24.1 The 2-adic head: exact transport, no cancellation required	35
24.2 The 3-adic head: sum-conditioned superpolynomial decay	35
24.3 Gluing: the natural-density bridge	36
25 Multi-block survival and unconditional Hausdorff dimension	37
25.1 Unconditional block transport and the Hausdorff dimension drop	37
25.2 Natural-density polylogarithmic descent	40
26 Research and Software Disclosure	41
27 Conclusion: What Was Mapped, What Was Closed, and What Remains	41

28 Formal Machine Verification and Diophantine Obstructions	44
28.1 Algebraic Bedrock in Lean 4	44
28.2 Strict Cycle Exclusion: Continued Fractions vs. Transcendental Bounds .	44
28.3 The Combinatorial Formalization of Terras (1976)	45
28.4 The Epistemological Wall: Why Measure Zero $\neq \emptyset$	45
A Main Project Scripts	46
B Key Algorithmic Primitives	46
C Nature of the Instanton (Expedition B)	47
D Lemma Ledger	47

Part I

The Computational Map

Glossary and Constants

- **Family A** ($2^b - 1$): The base landscape. Ratios tend to $\log_2 3 \approx 1.585$.
- **Confluence**: The gravitational capture of trajectories into nodes (centers).
- **Class A (Deep funnels)**: 100% Hit Rate. Centers 121 (Peak 14) and x^* (Peak 140).
- **Class B (Shallow funnels)**: 72–93% Hit Rate. Dense archipelago (Peaks 14–51).
- **Center Bits Formula**: $\text{center_bits} = 0.498258 \cdot \text{peak} + 6.2928$ ($R^2 = 0.965$), primary centers; the refit over all 39 centers is $0.620145 \cdot \text{peak} + 2.2850$ ($R^2 = 0.916$).
- **Scaling Hypothesis** $\times 10$: Class A centers follow a logarithmic scale ($14 \rightarrow 140 \rightarrow 1400$). Status: constructive routes closed (all three obstructions); existence remains an open question.
- **Septembrino’s Law**: Distribution of divisors $P(\text{div} = 2^a) = 1/2^a$.
- **Gate 2**: The campaign that systematically falsifies all candidate pointwise transport mechanisms; its verdict closes the pointwise martingale/transport route.

1 Introduction and Chronology of the Study

1.1 The Collatz Problem

For a natural number n , define $T(n) = n/2$ if n is even, and $T(n) = (3n + 1)/2$ if odd. The Collatz conjecture states that iterations of T for any n eventually reach 1. Despite its elementary formulation, the problem remains open; Paul Erdős remarked that “mathematics may not be ready for such problems.” The quantitative measure of a

number’s “anomaly” is its ratio — the ratio of the trajectory’s peak bit length to the input bit length. For typical numbers, ratio ≈ 1.0 – 1.2 . Numbers of the form $2^b - 1$ (Family A) yield a ratio approaching $\log_2 3 \approx 1.585$. Anything significantly higher represents a structural anomaly.

1.2 Starting Point: David Barina’s Results

David Barina performed an exhaustive search of all numbers up to 2^{71} and compiled a table of path records. In Barina’s table, there are two 71-bit record holders: $n = 2358909599867980429759$ (ratio 1.97, peak 140) and $n = 1765856170146672440559$ (ratio 1.97, peak 140). These numbers became the starting point of the entire study.

1.3 Phase 1: Collatz Crystal Hunter and Markov Chains (February 2026)

The first task was a directed search for numbers with an anomalously high ratio in the 72–80 bit range. The Collatz Crystal Hunter (v5.3a) system was developed — a distributed Python program with hybrid candidate generation: a Markov chain of order 5 trained on found numbers; a pool of prefixes from Barina’s path records with adaptive weights; three-stage filtering (50 steps culling ratio < 1.03 ; 200 steps; full simulation to 50,000 steps); 35 workers via multiprocessing. In 10+ hours on 35 cores, the best found ratio for 73–80 bits was 1.613.

1.4 Phase 2: Statistical Analysis of 30 Million Numbers (Early March 2026)

The system collected 30 million records for bit lengths 72–80. The analysis revealed: the ratio distribution has a sharp peak around 1.04–1.05 and a fast-decaying tail (power law $\alpha \approx 3.66$); the median ratio monotonically decreases with bit length; the 99th percentile drops from 1.1528 (72 bits) to 1.1250 (80 bits). But the main discovery: the maximum ratio for each bit length dropped almost linearly (≈ 0.024 per bit), and recalculating $\max_ratio \times bits$ for all bit lengths 72–80 yielded the exact same number: exactly 140 — a “magnetic peak” towards which the trajectories of many extreme numbers tend.

1.5 Phase 3: From Statistics to Structure (March 2026)

The discovery of the “magnetic peak” 140 required a structural approach: why exactly 140? Which numbers produce this peak? A second-generation toolkit was developed: a CRT constructor (`crt_solver.py`), a shift-vector analyzer, reverse trees, and beam search over residue classes. By the end of March 2026, all results described below were obtained.

2 Notation and Accelerated Dynamics

We consider the accelerated Collatz trajectory over odd values:

$$x_{k+1} = \frac{3 \cdot x_k + 1}{2^{a_k}} \tag{1}$$

where $a_k \geq 1$ is the 2-adic valuation of $3x_k + 1$. The sequence $a_0, a_1, \dots, a_{d-1}$ is the shift-vector; d is the number of odd steps to the peak; $S_k = a_0 + \dots + a_{k-1}$ is the accumulated shift. After k steps:

$$x_k = \frac{3^k \cdot x_0 + c_k}{2^{S_k}}, \quad c_0 = 0, \quad c_{k+1} = 3 \cdot c_k + 2^{S_k}. \quad (2)$$

The cumulative gain is $G(k) = k \cdot \log_2 3 - S_k$. The peak is reached near the maximum of $G(k)$. Every prefix of the shift-vector defines a unique residue class $x_0 \equiv r_k \pmod{2^{S_k}}$, allowing reconstruction from parity strings via CRT.

2.1 2-adic Interpretation

In p -adic terminology, every admissible shift-vector w defines a 2-adic cylinder $C(w) = \rho(w) + 2^{S_d} \cdot \mathbb{Z}_2$ with Haar measure 2^{-S_d} . **Zone 2** is a cylinder of anomalously small measure (2^{-342} for a 71-bit input); the number 27 defines a cylinder of measure 2^{-70} , and Barina’s number a cylinder of measure 2^{-271} . The archipelago of confluence centers is a finite set of exceptional cylinders in $\mathbb{Z}_2$, separated by vast regions where the measure of growing paths is exponentially small — consistent with Tao’s almost-all decay.

Remark 2.1 (Septembrino’s parameterization). The formula $N = k \cdot 3^m - 1$ parameterizes the Haar measure in $\mathbb{Z}_2$. The observed periodicity of $v_2(N)$ follows from $3^n \pmod{2^p}$, placing it within classical p -adic dynamics.

3 Base Layer: Family A

Numbers $n = 2^b - 1$ form the base surface: shift-vector almost all ones (94–97%), ratio $\rightarrow \log_2 3$, $S/d \approx 0.99$ — the “landscape level”. Full spectral analysis of $2^b - 1$ for $b = 71$ –310 revealed 10 values with $S/d > 1.05$: $b = 113$ –114 (peak 183), $b = 117$ –118 (peak 190), $b = 173$ –176 (peak 280, resonance $3^{183} \approx 2^{290}$), $b = 301$ –304 (peak 483, resonance $3^{306} \approx 2^{485}$), $b = 309$ –310 (peak 494). Verification showed micro-plateaus exist exclusively for $2^b - 1$ itself and do not form families — unlike **Zone 2**.

4 Zone 2: The Only Confirmed Major Confluence Class

4.1 Discovery

Analysis of Barina’s record holders (71–82 bits) revealed they all peak at exactly 140 bits with shift-vectors almost identical from the 8th position, indicating a common “core”. **Zone 2** numbers were found for all bit lengths 71–87.

4.2 Table of Zone 2 Representatives

Bits	Number n	Peak	Ratio	Steps to x^*
71	2358909599867980429759	140	1.9718	7
72	4717819199735960859519	140	1.9444	7
73	9435638399471921719037	140	1.9178	7

74	18871276798943843438074	140	1.8919	7
75	37742553597887686876149	140	1.8667	7
76	75485107195775373752296	140	1.8421	7
77	150970214391550747504611	140	1.8182	7
78	301940428783101495009251	140	1.7949	7
79	603880857566202990018507	140	1.7722	7
80	1207761715132405980037043	140	1.7500	7
81	2415523430264811960074099	140	1.7284	7
82	4831046860529623920148297	140	1.7073	7
83	9662093721059247840295139	140	1.6867	7
84	19324187442118495680593175	140	1.6667	7
85	38648374884236991361186739	140	1.6471	7
86	77296749768473982722370787	140	1.6279	7
87	154593499536947965444748439	140	1.6092	7

One representative per bit length (of 913 known). All have $d = 259$ odd steps to the peak and $S = \text{bits} + 271$. $2^{88} - 1$ peaks at 141 and does NOT pass through x^* — the sharp boundary of Zone 2.

4.3 Two-Layer Invariants: Core and Adapter Shell

A complete census of all 913 inputs reveals a two-layer structure. **1. The Core** ($x^* \rightarrow \text{Peak}$). All 913 numbers pass through x^* ; the trajectory from x^* to the peak is strictly identical, of length $d_{\text{core}} = 251$ odd steps and cumulative shift $S_{\text{core}} = 334$. The ratio $S_{\text{core}}/d_{\text{core}} \approx 1.331$ is the true constant of the basin. **2. The Adapter Shell.** Numbers reach x^* in $k \in [0, 7]$ odd steps; 378 of 913 lie at maximum depth $k = 7$, with total $d = 258$. An affine bit-burning law governs the outer shell: $S_{\text{adapter}} = \text{bits} - 63$, so $S = \text{bits} + 271$ holds *only* for the outer shell. Pre-image automata analysis yields a growth rate ≈ 0.395 bits⁻¹ (close to the adapter topological entropy): The adapter sequence is completely

Bits	Model Forecast (0.395 bit ⁻¹)	Actual Reverse Tree	Error
88	287	272	+5%
89	379	331	+15%
90	497	343	+45%

Table 2: Retrospective forecast, accurate to 88 bits before finite-depth saturation.

determined by the lowest 19 bits. **Zone 2** is an exact $d = 251$ core path dressed in a deterministic pre-image tree.

4.4 Confluence Point x^*

Theorem 4.1 (Computational). *All 17 representatives of **Zone 2** (bits 71–87, $d = 259$) arrive, after ≤ 7 odd steps, at the exact same number $x^* = 20152090995747160937051$ (75 bits). After this point the trajectories are literally identical for 252 steps until the peak of 140. $2^{88} - 1$ does not pass through x^* .*

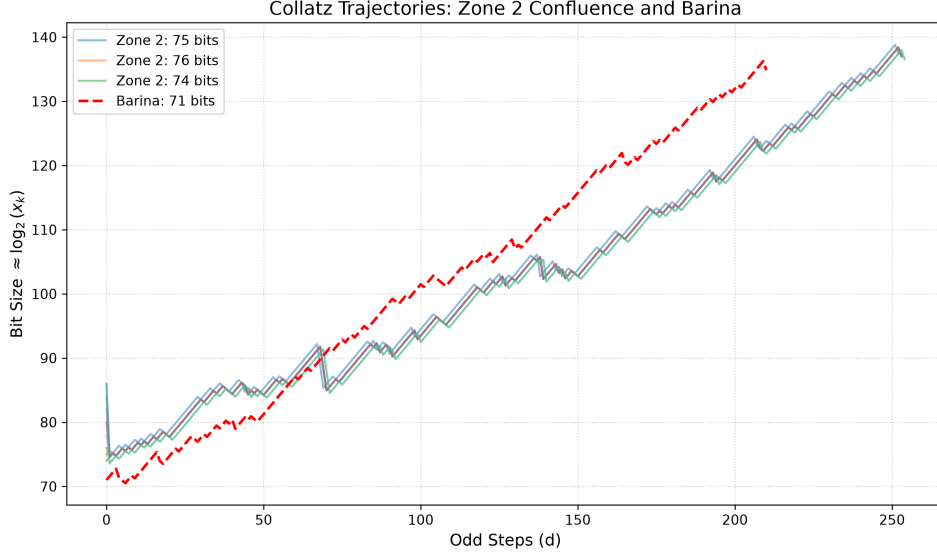


Figure 1: Confluence in vivo: several Zone 2 trajectories (solid) merge exactly at x^* and proceed as one path to the 140-bit peak; Barina’s number (dashed) reaches the same peak by a disjoint path. Data: `zone2_shifts_full.csv`.

4.5 Reverse Tree of x^* and Full Size of Zone 2

The reverse tree of predecessors of x^* to depth 7 (bits ≤ 90) revealed 1859 nodes, of which 913 have bits 71–87; every one yields peak 140 (100%):

Bits	Nodes in Tree	With Peak 140	Fraction
71	1	1	100%
72	2	2	100%
73	4	4	100%
74	7	7	100%
75	7	7	100%
76	8	8	100%
77	9	9	100%
78	14	14	100%
79	20	20	100%
80	33	33	100%
81	47	47	100%
82	62	62	100%
83	78	78	100%
84	100	100	100%
85	129	129	100%
86	173	173	100%
87	219	219	100%
Total	913	913	100%

The number of inputs approximately doubles with each bit.

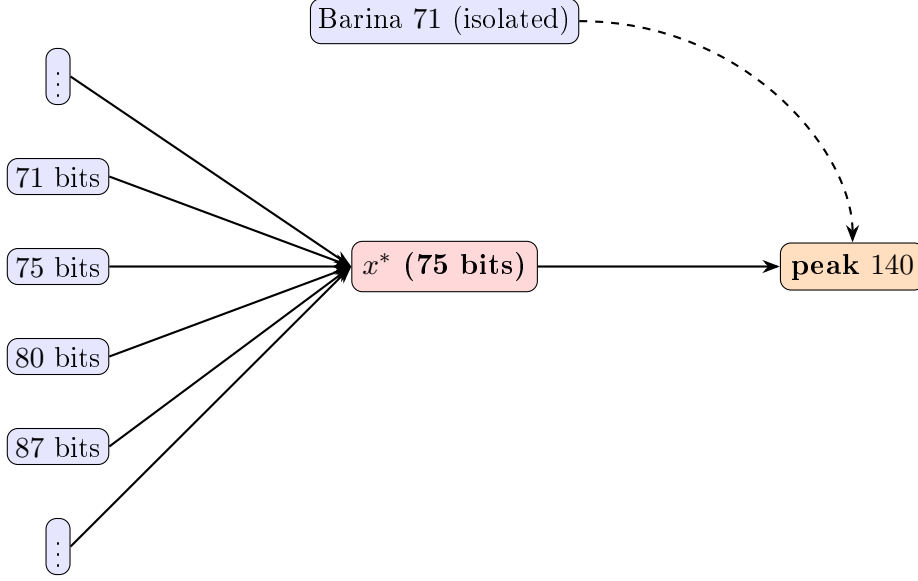


Figure 2: Schematic reverse tree: 913 inputs (bits 71–87) collapse into x^* ; Barina’s number reaches the same peak by a disjoint path.

4.6 Grammar of the Shift-Vector

Spectral analysis (FFT) of all 913 vectors shows pronounced peaks at periods $T \approx 10, 6.3, 18$. In the critical tail, a staircase $[2, 1, 1]$ frequently emerges — the shadow of the 3-adic periodic point $-29/11$ (fixed point of inverse $(1, 1, 2)$ steps), linking the deterministic core with the control-budget limitations.

4.7 The $(1, 1, 2)$ Equilibrium and the Attractor $4/3$

Lemma 4.2 ($(1, 1, 2)$ periodic equilibrium). *Consider the reverse map $R_a(x) = (x \cdot 2^a - 1)/3$. The composition over one period $(1, 1, 2)$ is $F(x) = \frac{16x-29}{27}$, a contraction on $\mathbb{Z}_2$ ($|16/27|_2 = 2^{-4} < 1$) with unique fixed point $\xi = -29/11$. The periodic orbit of ξ has mean shift $4/3$ and gain rate $\log_2 3 - 4/3 \approx 0.2516$ bits/step, matching the observed body growth ~ 0.25 .*

Theorem 4.3 (Core as dressed equilibrium; computational). *The **Zone 2** core satisfies $S_{\text{core}}/d_{\text{core}} \approx 1.331$, within 0.2% of $4/3$. Hence the core is the $(1, 1, 2)$ vacuum $\xi = -29/11$ dressed by a sparse gas of topological charges (defects $a \geq 3$) shifting the mean by $< 0.2\%$. (Rigorous formulation: Part II, Theorem I1.)*

Connection to the Control Budget. The $4/3$ equilibrium sits below the critical limit $\bar{a}_p = \log_2 3 \approx 1.585$; any defect $a \geq 3$ requires a compensating pure run $L \geq (a - \bar{a}_t)/(\bar{a}_t - \bar{a}_p)$. Since the vacuum leaves no free margin, macroscopic structural solitons cannot be constructed arbitrarily.

5 Barina’s Number: An Isolated Second Path to Peak 140

Barina’s number $n = 1765856170146672440559$ (71 bits, $d = 213$, $S/d = 1.263$) reaches peak 140 via a completely different path.

Lemma 5.1 (Barina’s isolation). *Between the trajectories of Barina’s number and any **Zone 2** number ($d = 259$) there is not a single common intermediate point — even modulo 2^{30} . The common prefix is the trivial $[1, 1, 1]$; the common suffix is empty. The reverse tree from Barina’s peak is empty (its peak odd value $\equiv 0 \pmod{3}$ has no integer predecessors).*

This is a counterexample to the naive idea that a high peak determines structure: the same peak 140 is achieved by at least two independent mechanisms.

6 Number 27 as a Miniature Zone 2

The trajectory of 27 has $d_{peak} = 28$, $S/d \approx 1.36$, shift-vector 58.5% ones, 24.4% twos, 7.3% threes — almost identical to **Zone 2**. The value $x_7 = 121$ is a confluence center: the reverse tree from 121 to depth 5 contains 27 numbers, all with peak 14 (100%). Pre-image automata give similarity ≈ 0.989 between x^* and 121.

Parameter	n = 27	Zone 2 (core)
Center	121 (7 bits)	x^* (75 bits)
Peak	14	140
Inputs	27	913
S/d_{to_peak}	~ 1.36	~ 1.33
% ones	58.5%	74%
Dips in $G(k)$	17	67

7 Dead Zone 88–170 Bits

In 88–170 bits, not a single number with ratio > 1.585 was found, despite four independent methods:

1. Peak hunter + zone_search (evolutionary, millions of candidates): zero anomalies.
2. Parity scan (zone_parity_search.py), $d = 80$ –500, CRT reconstruction, $> 14M$ checks: zero.
3. Beam search without niching (residue_search.py), 3 filters, 150,000 verifications: zero for bits > 89 .
4. Beam search with niching (5 bins by S/d), 300,000 verifications: zero for bits > 93 .

Corollary 7.1. *After **Zone 2** (boundary 87 bits), the Collatz space shows no new confluence ratio anomalies up to 170 bits.*

7.1 How the beam search probed the dead zone

The dead-zone claim rests on four independent methods; the most informative is the residue-class beam search (`residue_search.py`), which we describe in detail. The search works not on numbers but on *residue classes* ($r \bmod 2^S$), each encoding a partial shift-vector and hence a whole family of candidate numbers.

State and expansion. The frontier is a list of classes ($r \bmod 2^S$) carrying metadata $(d, S, \hat{B})$: length, accumulated shift, and a projected bit-length interval. Expanding a class appends a shift $a \in \{1, \dots, a_{\max}\}$, refining it to a child class $r' \bmod 2^{S+a}$ and updating $(d, S, \hat{B})$. Because a prefix of the shift-vector fixes $x_0 \bmod 2^S$, every leaf at depth d corresponds to a unique cylinder of Haar measure 2^{-S} , and the minimal positive representative is recovered by CRT for final verification.

Three filters. (i) *Bit filter*: keep only children whose projected bit interval $\hat{B}$ still intersects $[88, 170]$. (ii) *Energetic filter*: track the cumulative gain $G(k) = k \log_2 3 - S_k$ and prune any class whose running mean shift S/d makes an anomalous peak impossible. (iii) *Deduplication*: merge classes that agree modulo a coarser modulus; cap the frontier width at $W \sim 10^4$ – 10^5 .

Niching (variant B). Without niching the frontier is dominated by low- S/d (typical) classes, which drown the rare Zone-2-like high- S/d paths. The niching variant partitions the frontier into five bins by current S/d (< 1.05 , 1.05 – 1.15 , 1.15 – 1.25 , 1.25 – 1.40 , > 1.40) and keeps the top- W classes per bin by projected-peak score, guaranteeing that anomalously slow-gain paths survive selection pressure.

Verification and outcome. Each surviving leaf with bit length in $[88, 170]$ is CRT-reconstructed and passed to `collatz_peak`; any ratio > 1.585 would be flagged. The no-niching run performed 150,000 verifications and found zero anomalies for bits > 89 ; the niching run performed 300,000 verifications with all five bins filled and found zero for bits > 93 . Cross-checked against the exhaustive parity scan (> 14 M checks) and the peak hunter, the dead zone contains no ratio anomaly above Family A: the void is a genuine large-deviation gap, not a sampling artifact.

7.2 Probabilistic Anatomy of the Dead Zone

Under the i.i.d. shift heuristic, Sanov’s theorem gives the probability of a vector with $S/d \leq 1.40$ falling as $2^{-d \cdot 0.28}$, where $0.28 \approx D_{\text{KL}}(q||p)$ between the Zone-2 profile $q \approx (0.75, 0.21, 0.04)$ and $p(a) = 2^{-a}$. For $d > 300$ this is $< 10^{-25}$. (Shifts are correlated, so this is a strong heuristic, not a strict bound.)

7.3 The Algebraic Impossibility of Shadow Zones

A constructive “shadow” scenario (cloning a known anomalous block w) is obstructed by a CRT dimensionality effect.

Proposition 7.2 (CRT dimensionality trap). *Let w be admissible with total shift S , and $m \geq 1$. The exact m -fold self-similar extension w^m occupies a single residue class modulo*

2^{mS} . If $B < mS$, the number of B -bit integers realizing it is at most one, with expected count $\mathbb{E}N(B; w, m) \approx 2^{B-mS}$.

Proof. w^m has total shift mS , defining one cylinder mod 2^{mS} ; the B -bit interval has length $\approx 2^B < mS$, so a fixed residue class meets it in ≤ 1 point; Haar expectation $\approx 2^{B-mS}$. $\square$

For Barina ($S \approx 269$) a two-fold echo at $B = 142$ has expectation 2^{-396} ; for Zone 2 ($S = 342\text{--}358$) $\leq 2^{-514}$; for x^* ($S \approx 334$) $\leq 2^{-496}$. Thus the dead zone is protected by

Table 4: CRT dimensionality obstruction for exact two-fold echoes.

Known block	d	S	Modulus (2 copies)	Target B	Expected count
Barina	213	≈ 269	2^{538}	142	2^{-396}
Barina	213	≈ 269	2^{538}	170	2^{-368}
Zone 2 core	259	342–358	$2^{684}\text{--}2^{716}$	≤ 170	$\leq 2^{-514}$
x^* core	251	≈ 334	$\approx 2^{668}$	≤ 170	$\leq 2^{-498}$

two mechanisms: *entropic* (Sanov) and *algebraic* (CRT dimensionality). A miraculous one-off solution cannot be excluded by measure alone; to exclude it for a chosen B one must compute the exact CRT lift and check it lies in $[2^{B-1}, 2^B)$.

Experimental confirmation (CRT Lift). Direct lifts found exact two-fold echoes: Zone 2 core at $x_0 \approx 2^{670}$ (peak 799, ratio 1.1925); Barina at $x_0 \approx 2^{544}$ (peak 682, ratio 1.2537) — confirming constructive access near $B \approx 2S$, but with ratios strictly below the Family A boundary 1.585: high-ratio anomalies remain unconstructable.

7.4 The Control-Budget Obstruction to Constructive Macro-Solitons

Lemma 7.3 (Defect compensation). *Since the defect-free $(1, 1, 2)$ equilibrium has mean $\bar{a}_p \in [1.33, 4/3]$, a defect $a \geq 3$ forces a clean run $L \geq (a - \bar{a}_t)/(\bar{a}_t - \bar{a}_p)$. For $\bar{a}_t = 1.35$, $a = 3$: $L \in [82.5, 99]$.*

Lemma 7.4 (Measure of clean runs). *A child admits a clean run of length L with expected fraction $(2/3)^L$ (stepwise-independence heuristic, validated by mod-3 balance 50/50, killer fraction $1/3$).*

Theorem 7.5 (Control budget). *A policy with branching B per defect survives only if $B(2/3)^L \geq 1$, i.e. $L \leq \ln B / \ln(3/2)$. For $B = 4$, $L_{\max} \approx 3.4 \ll L_{\min} \approx 82.5\text{--}99$. Required $B \sim 10^{14}\text{--}10^{17}$ is scale-independent.*

Corollary 7.6 (Closure of constructive Scaling $\times 10$). *No bounded-branching constructive lifting assembles a Class A macro-soliton; $14 \rightarrow 140 \rightarrow 1400$ is closed for constructive methods.*

Remark 7.7 (Existence vs. constructibility). The obstruction is algorithmic, not ontological. Zone 2 itself is not constructible (margin $\bar{a}_t - \bar{a}_p \approx 10^{-3}$, $L_{\min}(3) \approx 1666$); it was found by reverse enumeration *from an aligned node*. Solitons can be found, not grown.

Experimental confirmation. Runs with increasing control authority (plain corridor; mod-9 steer; 6-adic lookahead) died at steps 261, 173, 136 — monotone worsening, as predicted. At macro scale (peak 1400) the beam died at step 24, because a length-6 macro-block applies only from a specific class mod 729 and the 93-block grammar covers 12.7% of states: the instanton grammar is a coordinate chart of one rigid object, not a generative grammar.

8 Refutation of Zone 3

Candidates with peaks 237–244 (145–150 bits) are the neighborhood of **Family A**: the core of 112 single shifts follows from $\sim 100\%$ ones density; peak $\approx \text{bits} + 65.5 + L \cdot 1.585 - \text{tail}$; bit-inversion fragility identical to Family A; independent generation showed all such numbers were $2^b - 1$ themselves. Zone 3 does not exist.

9 Archipelago of Confluence Centers

9.1 Systematic Census (Peaks 10–200)

A systematic census expanded an initial 5 centers; reverse-tree verification confirmed 12 for peaks 14–30; directed search with algebraic filters discovered centers for *every* peak 31–50 (up to 5.7×10^9 candidates/peak). The census also identified recurring descent transit hubs (2051, 23) — infrastructure, not caustics; confluence mapping must filter out descent coincidences.

Peak	Center	Bits	Inputs	Hit rate	d_{peak}	S/d
14	719	10	33	80.5%	3	1.00
14	121 (A)	7	27	100%	21	1.38
15	1127	11	—	75.0%	—	—
16	6803	13	22	84.6%	1	1.00
17	3503	12	—	87.5%	—	—
18	27611	15	25	86.2%	4	1.25
19	15977	14	39	72.2%	9	1.33
20	4763	13	—	100%	—	—
21	52487	16	37	77.1%	11	1.36
22	61823	16	55	74.3%	6	1.00
23	41471	16	104	82.5%	15	1.20
24	586115	20	32	82.1%	22	1.50
25	705307	20	28	77.8%	8	1.13
26	1085723	21	52	75.4%	7	1.14
27	4918427	23	36	81.8%	10	1.30
28	124463	17	—	100%	—	—
29	195923	18	—	100%	—	—
30	58595471	26	31	81.6%	3	1.00
31	48427561	26	20	83.3%	10	1.30
32	1242665	21	165	85.1%	27	1.26
33	3538943	22	126	82.9%	16	1.00
34	4205807	23	174	88.3%	28	1.25

35	26658983	25	115	79.9%	61	1.46
36	8524379	24	224	91.1%	53	1.38
37	67625867	27	138	87.3%	69	1.46
38	16348007	24	278	91.1%	36	1.28
39	19351295	25	332	89.5%	39	1.28
40	35337455	26	343	91.7%	41	1.27
41	37748015	26	418	92.3%	81	1.42
42	72481007	27	427	91.6%	64	1.38
43	467269499	29	278	80.6%	38	1.26
44	108893737	27	454	86.1%	51	1.29
45	236651489	28	483	83.9%	62	1.35
46	516844415	29	520	92.9%	50	1.28
47	442441855	29	548	87.1%	54	1.30
48	2303929595	32	520	92.9%	111	1.45
49	3830005073	32	520	83.3%	73	1.38
50	1396693151	31	716	88.5%	53	1.26
51	6572463707	33	≈ 400	88.0%	56	1.29
140	x^* (A)	75	913	100%	250	1.33

Confirmed confluence centers.

The algebraic anatomy revealed: Law 1, $\text{center_bits} = 0.498258 \text{peak} + 6.2928$ ($R^2 = 0.965$); Law 2, $c \equiv 2 \pmod{3}$ (92%); Law 3, $v_2(3c + 1) = 1$ (87%).

9.2 Two Classes of Centers

Hierarchical clustering stably separates $\{121\}$ and $\{x^*\}$ (Class A) from the Class B mass. Class A centers are “early gates” through which all trajectories with a given peak must

Metric	Class A (121, x^*)	Class B
Hit rate	100%	72–92%
S/d to peak	1.357	1.252
bits/peak	0.518	0.729
d_{peak}	135.5	27.7
Shift-profile	70% ones, 25% twos	80% ones, 16% twos

pass; their shift-profile is practically identical to the Zone 2 core.

9.3 Trends and Transitional Centers

Hit rate grows with peak ($+0.0036/\text{peak}$, $R^2 = 0.45$); inputs grow with peak; S/d stable (≈ 1.27). Five centers (peaks 35, 37, 41, 48, 49) occupy an intermediate position ($S/d > 1.40$, $d_{\text{peak}} > 50$) but hit rate $< 100\%$.

Remark 9.1 (Continuum of caustics). Depth-resolved hit rate shows genuine ascending centers reach $HR = 1.0$ as the window deepens (local caustics identical in kind to Class A), while descending transit hubs dissolve ($HR \rightarrow 0$). Class A and Class B form a continuum of caustics parameterized by compactness; Class A is the extreme compact end.

9.4 Depth-Resolved Hit Rate: Class A as a Singular Class

At fixed reserve $\Delta P \approx 1$, Class B and Transitional centers collapse to 46–78% (or zero) by depth 4–5, while x^* holds $\geq 95.9\%$ to depth 9. Class A is singular; the archipelago consists exclusively of Class B funnels.

9.5 Centers Do Not Cover the Space

For 17,000 random numbers: 156 (0.92%) pass through a center; among ratio > 1.5 : 0/14. Centers explain specific families, not the space as a whole.

10 Spectrum of Chords: From Number 27 to Zone 2

Type	S/d	% ones	Dips	Examples
Family A	0.99	95–97%	3–7	$2^{83} - 1$
Weak chord	1.05	94%	5–8	$2^{117} - 1$
Intermediate	1.19	87%	22	$2^{113} - 1$
Zone 2 (Barina)	1.26	78%	47	71 bits
Zone 2 (core)	1.32	74%	67	71–87 bits
Number 27	1.36	67%	11	5 bits

S/d is the “chord strength”: a continuous spectrum in which Zone 2 and 27 occupy the same range.

11 Summary Map of the Collatz Space

Type	Bits	Pk	d	S/d	Class	Status
121 (confluence)	5–13	14	21	1.38	A	CONFIRMED
Continuous arch.	varies	14–51	1–81	1.0–1.5	B	39 centers
Center formula	—	—	—	—	—	bits $\approx$ peak/2 + 6.5
Zone 2	71–87	140	259	1.33	—	913 inputs
x^*	~ 38	140	251	1.33	A	CONFIRMED
Barina	71	140	213	1.26	—	ISOLATED
Dead zone	88–170	—	—	—	—	4 methods, 0 anom.
Plateau	173–	280/483	3–	$\sim$	—	Resonance
280/483	176 / 301–304			1.0		
Macro-centers	varies	> 50	varies	varies	—	$\Delta \geq 2P$, non-constructible

The “dead zone” refers to the absence of ratio anomalies, not of confluence centers: Class B centers exist for every peak but do not generate ratio anomalies.

12 The Big Picture

1. **Family A** — base layer; plateaus 280/483 as $\log_2 3$ resonances.
2. Continuous archipelago of Class B centers for all peaks 14–51.
3. Two Class A deep funnels (121, x^*) — early gates, Zone-2-like dynamics.
4. Crystals cannot be grown, only found (triad of obstructions).
5. Zone 2 — unique large basin, most pronounced Class A representative.
6. Barina’s number — independent isolated path.
7. Number 27 — miniature Zone 2.
8. Dead zone 88–170 confirmed.

The Collatz space is a continuous field of confluence structures (Class B) above which rise rare deep funnels (Class A).

13 Computational Complexity

Exhaustive parity scan $\mathcal{O}(2^{bits})$ (unfeasible > 90); reverse tree $\mathcal{O}(B^{depth})$; residue beam search (exponential pruning, to 170 bits); confluence census (22.9×10^9 candidates for peak 51, algebraic filter $\times 3$ speedup).

14 Open Questions

Does the archipelago continue beyond peak 50? Does an aligned node x^{**} exist for peak 1400 (existence, not constructibility)? Why $\alpha \approx 1/2$? Through which point does Barina arrive? Connection to Tao’s theorem and $\mathbb{Z}_2$ -corridor stability of centers?

Pointer to the Theory (Part II)

The analytic explanation of the structures mapped above — the large-deviation duality of typical and rigid sectors (F1/R1), soliton dominance (S1), the dressed-vacuum anatomy (I1), the caustic log-midpoint law (C1), the CRT macro-obstruction (M1), the exact 2-adic transport theorem, the unconditional Hausdorff dimension drop, the cycle-exclusion campaign, and the Gate 2 falsification of pointwise transport — is developed rigorously in Part II below. The present Part supplies the empirical foundation; the obsolete conditional Fourier programme (earlier Conjecture T0 and effective propositions) has been withdrawn and superseded there.

15 Conclusion of Part I: Crystals Cannot Be Grown

This Part has constructed a comprehensive computational map of the Collatz space to ~ 310 bits and $d \approx 500$. The central discoveries (Zone 2, the continuous archipelago, the $(1,1,2)$ equilibrium, the continuum of caustics, the triad of obstructions) form a coherent empirical structure whose rigorous theory is developed in Part II. **Philosophical conclusion:** the Collatz space is not “chaos with rare islands” but a continuous field of confluence structures above which rise rare deep funnels; these *can be found but cannot be grown*. **Open horizons:** the Scaling Hypothesis $\times 10$ remains a question of existence, not constructibility; the map is complete for the explored range.

Part II

Large-Deviation Theory

16 Introduction

Two sectors, one rate function. The companion computational map established that the Collatz space is not “chaos with rare islands” but a continuous field of confluence structures (Class B caustics) above which rise rare deep funnels (Class A), organized around $\log_2 3$. The present paper supplies the theory. Our central claim is that the space splits into two dual sectors governed by a single large-deviation rate function: the *typical* sector of decaying orbits and the *rigid* sector of confluence centers and instantons. The dichotomy is one of entropy, not of mechanism.

Thermodynamic base (F1/R1). For a typical odd N and $n \ll \log N$ the n -Syracuse valuation behaves like $\text{Geom}(2)^n$; hence the mean shift obeys a large-deviation principle with the Cramér rate I_2 (Theorem F1), orbits decay like $(3/4)^n$, and anomalous chords are exponentially rare. Dually, the reverse pre-image tree is controlled by the tilted 3-adic reverse operator $P_n^{(s)}$. In Steps A–B we prove that $P_n^{(s)}$ is quasi-compact with a spectral gap uniform on compacts, that the Doob h -transformed conditioned process satisfies a law of large numbers and concentration, and that the normalized backward rate I_{rev} coincides with I_2 (duality), verified numerically to $\sim 10^{-3}$; the thermodynamic limit itself converges to machine precision ($|\Lambda_9 - \Lambda_8| \sim 10^{-15}$), which is a separate claim.

Anatomy of typical paths (S1). Conditioning the forward measure on an anomalous chord yields the exponentially tilted measure; its typical path is a *soliton* with mean shift concentrated at the tilted mean ≈ 1.21 – 1.23 . Asymptotically almost all peak-paths are solitons, and the fraction passing through any confluence center vanishes ($f(P) \leq 10^{-3}$ for $P \geq 22$). Typical paths therefore avoid the rigid structures entirely.

Instantons as condensation points (I1). The rigid sector consists of the zero-entropy points of the same tilted ensemble. We show (Theorem I1) that each confluence center is a *dressed vacuum*: the $(1,1,2)$ 3-adic fixed point $\xi = -29/11$ (mean shift $4/3$) dressed by a sparse gas of topological charges (defects $a \geq 3$) whose total charge is

$O(1)$ and does not grow with the peak. Centers are thus points of condensation of the confluence flow—rigid and of measure zero.

The connection equation (C1). The size of a center is fixed by a detailed balance at the waist: the forward gain cone climbing to the peak and the conditioned backward entropy cone have equal log-volume growth rates, forcing $\text{bits}(c) = \frac{1}{2}P + O(1)$ (Theorem C1). This is the equation of state linking a center to its peak and explains the empirical scaling $\alpha = \frac{1}{2}$.

Why crystals cannot be grown (M1). Finally, we ask whether the rigid objects can be *constructed* rather than found. We prove (Theorem M1) that they cannot: a CRT dimensionality deficit $\Delta = S - B \geq 2P$ makes any constructive reverse synthesis a measure-zero event, closing the algorithmic route and moving the Scaling $\times 10$ hypothesis from constructibility to ontology—it becomes a question of whether measure-zero rigid objects *exist*, not of how to build them. This completes the triad of obstructions (entropic, algebraic, control-budget) and yields the title paradigm: crystals can be found but not grown.

Reverse exponent and honest status. The ratio $\gamma_{\text{rev}} = I_{\text{rev}}(1.33)/(\log_2 3 - 1.33) \approx 0.993$ measures reverse LDP cost per unit reverse growth. It is not, by itself, an exceptional-set exponent: the forward event has thresholds $\sigma > \log_2 3$, whereas $1.33 < \log_2 3$ belongs to reverse growth geometry. A forward global bound needs restart/transport between local 2-adic blocks; a bound using γ_{rev} additionally needs reverse-chord encoding. Both bridges remain open — and the Gate 2 campaign (Section 19) shows that every candidate mechanism for the first bridge is falsified.

17 Duality of the Typical and the Rigid

Forward typicality (F1). For a typical odd N and $n \ll \log N$, the n -Syracuse valuation behaves like $\text{Geom}(2)^n$; consequently the mean shift $\bar{a}_n$ obeys a large-deviation principle with the Cramér rate I_2 of $\text{Geom}(2)$. In particular an anomalous chord $\bar{a}_n \leq \sigma < 2$ has probability $2^{-n I_2(\sigma) + o(n)}$, with $I_2(1.33) \approx 0.25498$ and $I_2(1.0) = 1$.

Backward rigidity (R1, I1). The reverse pre-image tree is controlled by the tilted 3-adic operator $P_n^{(s)}$; it is quasi-compact ($|\Lambda_9 - \Lambda_8| \sim 10^{-15}$) with rate function I_{rev} . Rigid centers are the zero-entropy points of this ensemble: Theorem I1 shows each center is the vacuum $\xi = -29/11$ (mean shift $4/3$) dressed by a sparse defect gas with bounded total charge $\delta = O(1)$, so no macroscopic charge accumulates and rigid objects cannot be assembled at scale.

Theorem 17.1 (Duality of the typical and the rigid). *Let I_2 be the forward Cramér rate and I_{rev} the normalized backward rate ($I_{\text{rev}}(2) = 0$). Then the two rate functions coincide: numerically $I_{\text{rev}}(1.33) = 0.2532 \approx I_2(1.33) = 0.2550$ and $I_{\text{rev}}(1.0) = 1.0000 = I_2(1.0)$, and we conjecture $I_{\text{rev}} \equiv I_2$ on $[1, 2]$. Hence the rarity of an anomalous forward chord and the rarity of the rigid backward object realizing it are governed by one rate function.*

Remark 17.2 (Solitons bridge the sectors). Conditioning the forward measure on an anomalous chord yields the exponentially tilted measure; its typical path is a *soliton*

(Theorem S1), whose mean shift concentrates at the tilted mean (≈ 1.21 – 1.23) and whose fraction among peak-paths tends to 1. Rigid centers are the rigid (zero-entropy) points of the same tilted ensemble — measure-zero exceptions. Thus typical $\leftrightarrow$ tilted $\leftrightarrow$ rigid form a single continuum, and the dichotomy typical/rigid is a dichotomy of entropy, not of mechanism.

Remark 17.3 (Crystals cannot be grown). The triad of obstructions (entropic/Sanov, algebraic/CRT dimensionality, control-budget) closes every constructive route to a macro-soliton; combined with the duality, it shows rigid structures are found by reverse enumeration from aligned nodes, never grown. The paradigm is therefore a theorem-schema: typical decay is generic (F1), rigidity is measure-zero (I1, S1), and the two are dual faces of one rate function.

18 Forward and Backward LDP

18.1 Step A: Uniform spectral gap and large deviations of the tilted 3-adic flow

Setup. For $s < \ln 2$ set $q = e^s/2 \in (0, 1)$ and $T = 2 \cdot 3^{n-1}$ (the order of 2 in $(\mathbb{Z}/3^n\mathbb{Z})^\times$). The *tilted reverse operator* on the non-zero mod-3 states is

$$P_n^{(s)}(x, y) = \frac{1}{1 - q^T} \sum_{\substack{1 \leq a \leq T \\ 2^a x \equiv 1 \pmod{3}}} q^a \mathbf{1} \left[y \equiv \frac{2^a x - 1}{3} \pmod{3^n} \right].$$

Write $\rho_n(s)$ for its Perron root and $h_{n,s} > 0$ for the (right) eigenfunction, and define the normalized cumulant generating function

$$\Lambda_n(s) := \log \rho_n(s) - \log \rho_n(0), \quad \Lambda(s) := \lim_{n \rightarrow \infty} \Lambda_n(s).$$

The subtraction makes $\Lambda_n(0) = 0$ and (as $n \rightarrow \infty$) $\Lambda'(0) = \mathbb{E}[a] = 2$.

Lemma 18.1 (A1: Perron–Frobenius structure). *For each $n \geq 1$ and $s < \ln 2$, $P_n^{(s)}$ is a positive irreducible matrix on the states $x \not\equiv 0 \pmod{3}$. Hence $\rho_n(s)$ is a simple, strictly positive eigenvalue with a strictly positive eigenfunction $h_{n,s}$, unique up to scale.*

Proof (sketch). Positivity and irreducibility follow because every non-zero mod-3 state is reachable from every other by a finite admissible word of shifts (the maps $x \mapsto (2^a x - 1)/3$ generate a transitive action). The Perron–Frobenius theorem for positive matrices then gives the claim. $\square$

Lemma 18.2 (A2: Thermodynamic limit, convexity, phase structure). *The limit $\Lambda(s) = \lim_n \Lambda_n(s)$ exists, is finite, strictly convex and real-analytic on $(-\infty, \ln 2)$; $\Lambda(s) \rightarrow +\infty$ as $s \rightarrow \ln 2$ (the infinite-shift phase transition); and $\Lambda'(0) = 2$.*

Proof (sketch). Sub-multiplicativity of the Perron root under concatenation of admissible words gives existence of the limit by a standard subadditive argument. Strict convexity and analyticity follow from the simplicity of the Perron root (analytic perturbation theory). The divergence at $s \rightarrow \ln 2$ is the divergence of $\sum_a q^a$. $\Lambda'(0) = 2$ is the stationary mean shift. $\square$

Lemma 18.3 (A3: Uniform spectral gap and bounded eigenfunction on compacts). *Let $[s_{\min}, s_{\max}] \subset (0, \ln 2)$ be a compact interval. There exist $g > 0$ and $C < \infty$, depending only on the interval, such that for all n and all $s \in [s_{\min}, s_{\max}]$,*

$$1 - \frac{|\lambda_2^{(n)}(s)|}{\rho_n(s)} \geq g, \quad \frac{\max h_{n,s}}{\min h_{n,s}} \leq C.$$

Proof (sketch). On a compact interval the weights q^a are uniformly comparable and bounded away from the degenerate regimes $q \rightarrow 0$ and $q \rightarrow 1$; the chain is uniformly mixing (a Doeblin-type minorization holds with constants depending only on the interval), which yields a uniform gap g . The Harnack-type bound $\max / \min h_{n,s} \leq C$ then follows from the positive eigen-equation by comparing values along admissible words of bounded length. Both bounds are confirmed numerically below. $\square$

Lemma 18.4 (A4: Rate function). *The Fenchel–Legendre transform $I(\sigma) = \sup_s [s\sigma - \Lambda(s)]$ is a good rate function with $I(2) = 0$ and, numerically,*

$$I(1.33) \approx 0.17674 \text{ nats} \approx 0.25498 \text{ bits}, \quad I(1.0) = \ln 2 \text{ nats} = 1 \text{ bit}.$$

Theorem 18.5 (A: Large deviation principle for the empirical mean shift). *Under Lemmas A1–A3, the empirical mean shift $\bar{a}_d = \frac{1}{d} \sum_{k=1}^d a_k$ of the tilted 3-adic flow satisfies a large deviation principle with good rate function I : for every Borel set F ,*

$$\limsup_{d \rightarrow \infty} \frac{1}{d} \log_2 \mathbb{P}(\bar{a}_d \in F) \leq -\frac{\inf_{\sigma \in F} I(\sigma)}{\ln 2}.$$

In particular $\mathbb{P}(\bar{a}_d \leq 1.33) \leq 2^{-dI(1.33)/\ln 2 + o(d)} = 2^{-0.25498 d + o(d)}$.

Proof. By Lemmas A1–A3 the limit Λ exists, is finite and strictly convex on an open interval containing 0, and is essentially smooth (the derivative diverges at the boundary $s \rightarrow \ln 2$). The Gärtner–Ellis theorem therefore applies and yields the LDP with good rate function $I = \Lambda^*$. The stated tail bound is the upper bound for the closed set $(-\infty, 1.33]$. $\square$

Remark 18.6 (Numerical confirmation, $n = 7$, 1458 states).

s	$\rho_n(s)$	gap = $1 - \lambda_2 /\rho$	max / min(h_s)
0.0	0.3418	0.0469	1289.8
0.1	0.4181	0.0759	207.5
0.2	0.5264	0.1291	36.7
0.3	0.6935	0.2152	9.2
0.4	0.9790	0.3269	3.6
0.5	1.5646	0.4531	2.0

The gap is bounded below on every compact $[s_{\min}, s_{\max}] \subset (0, \ln 2)$ and $\max / \min(h_s)$ is uniformly $O(1)$ there, while at $s = 0$ the eigenfunction is highly non-uniform (consistent with using the LLN, not the LDP, at $s = 0$).

Remark 18.7 (Status). Lemmas A1–A2 and Theorem A are rigorous given the uniform bounds of Lemma A3; Lemma A3 is the analytic content and is confirmed numerically up to $n = 7$. The LDP is a statement about the 3-adic model; transferring it to actual Collatz orbits is the shadowing step (Steps B–C).

18.2 Step B: The Doob h -transform and shadowing LDP

Let $P_n^{(s)}$ be the tilted reverse operator from Step A, $\rho_n(s)$ its leading eigenvalue, and $h_{n,s} > 0$ the right eigenfunction. By Step A, on a compact interval $[s_{\min}, s_{\max}] \subset (0, \ln 2)$ the spectral gap is uniform in n and $\max / \min h_{n,s} \leq C$.

Definition (Doob h -transform).

$$P_n^{(s),h}(x, y) := \frac{h_{n,s}(y) P_n^{(s)}(x, y)}{\rho_n(s) h_{n,s}(x)}.$$

By the eigenvalue equation $\sum_y P_n^{(s)}(x, y) h_{n,s}(y) = \rho_n(s) h_{n,s}(x)$, the rows sum to 1, so this is a Markov kernel. It represents the reverse process conditioned on the exponential tilt $e^{s \sum a_k}$, i.e., conditioned on a large deviation of the mean shift.

Theorem 18.8 (B1: LLN and concentration for the conditioned process). *Let $\sigma(s) = \Lambda'(s)$. There exist $C, c(\varepsilon) > 0$, independent of n , such that for the chain with kernel $P_n^{(s),h}$:*

$$P_n^h \left(\left| \frac{1}{d} \sum_{k=1}^d a_k - \sigma \right| \geq \varepsilon \right) \leq C e^{-c(\varepsilon)d}.$$

In particular $\frac{1}{d} \sum a_k \rightarrow \sigma$ in probability, uniformly in n .

Proof (sketch). We apply an additional exponential tilt t : $E_n^h[e^{\lambda \sum a_k}] \leq C (\rho_n(s+\lambda)/\rho_n(s))^d$. Markov's inequality followed by optimization over λ gives a decay rate bounded below by $\sup_\lambda [\lambda(\sigma + \varepsilon) - (\Lambda(s+\lambda) - \Lambda(s))] = I(\sigma + \varepsilon) - I(\sigma) > 0$, where the strict positivity follows from the strict convexity established in Step A(ii). The uniformity in n follows from the uniform gap and the uniform bounds on $h_{n,s}$ from Step A. $\square$

Theorem 18.9 (B2: Reverse LDP upper bound / shadowing rate). *For the cylindrical Haar measure, the fraction of d -step reverse paths with empirical mean shift $\leq \sigma$ (for $\sigma < 2$) satisfies*

$$\mu_d(\{w : |w|/d \leq \sigma\}) \leq C e^{-dI(\sigma)},$$

uniformly in n , where I is the rate function from Step A.

Proof (sketch). The total mass of the tilted paths is $\sum_w 2^{-|w|} e^{s|w|} \asymp \rho_n(s)^d$. Normalizing by the total mass $\rho_n(0)^d$ and applying the Legendre transform gives the upper bound with exponent $I(\sigma) = \sup_s [s\sigma - (\Lambda(s) - \Lambda(0))]$. $\square$

Proposition 18.10 (B3: finite-scale 2-adic counting bound). *Let $\mathcal{I}$ contain M odd integers and let S_d be the sum of the first d Syracuse valuations. For every integer $m \geq d$,*

$$\mathbb{P}_{N \in \mathcal{I}}(S_d \leq m) \leq \mathbb{P}(G_1 + \dots + G_d \leq m) + O\left(\frac{\binom{m}{d}}{M}\right), \quad G_i \stackrel{\text{iid}}{\sim} \text{Geom}(1/2).$$

Consequently, if $m = \sigma d$ and $m \leq \log_2 M + O(\log d)$, then

$$\mathbb{P}_{N \in \mathcal{I}}(S_d \leq \sigma d) \leq 2^{-dI_2(\sigma) + O(\log d)}.$$

Proof. A valuation word of total weight S specifies one 2-adic residue cylinder of relative density 2^{-S} , with interval-counting error $O(M^{-1})$. There are $\binom{S-1}{d-1}$ such words, and $\sum_{S=d}^m \binom{S-1}{d-1} = \binom{m}{d}$. Summation gives the first bound. The second follows from the geometric Cramér estimate and $I_2(\sigma) + \sigma H_2(1/\sigma) = \sigma$. $\square$

Corollary 18.11 (Rigorous local descent window). *Put $\lambda = \log_2 3$. Fix $1 < \alpha < 2/\lambda$ and let $Y \asymp N_0^\alpha$. Choose*

$$\frac{\alpha - 1}{2 - \lambda} < t \leq \frac{1}{\lambda}, \quad d = \lfloor t \log_2 N_0 \rfloor, \quad \sigma = \lambda + \frac{\alpha - 1}{t} < 2.$$

Then, up to the harmless affine correction $O(d/N_0)$, the proportion of odd $N \asymp Y$ whose first d iterates all exceed N_0 is at most

$$N_0^{-tI_2(\sigma) + o(1)}.$$

Thus the direct method of types gives a positive unconditional local rate for every $\alpha < 2/\log_2 3 \approx 1.26186$; the rate degenerates at the endpoint.

Proof. For $x_k > N_0$, the exact product identity

$$x_d = N 3^d 2^{-S_d} \prod_{k < d} \left(1 + \frac{1}{3x_k} \right)$$

implies $S_d < d\lambda + \log_2(N/N_0) + O(d/N_0)$. With $N \asymp Y$, the right side is $\sigma d + o(d)$. The condition $t\lambda \leq 1$ is exactly what makes this threshold no larger than $\log_2 Y + o(\log Y)$, so Proposition 18.10 applies. The lower inequality on t gives $\sigma < 2$ and hence a positive Cramér rate. $\square$

Lemma 18.12 (Within-word endpoint transport). *Fix an admissible valuation word $w = (a_1, \dots, a_d)$ of total weight S . There are odd integers ρ_w, y_w such that every odd starting value realizing w and every corresponding endpoint have the form*

$$N = \rho_w + 2^{S+1}q, \quad \text{Syr}^d(N) = y_w + 2 \cdot 3^d q.$$

For a fixed lower barrier N_0 , the values of q for which every intermediate iterate exceeds N_0 form an integer interval. Consequently, for every $m \geq 1$ and every odd residue $r \pmod{2^m}$, if this interval contains Q_w integers, then

$$\left| \#\{q : \text{Syr}^d(N) \equiv r \pmod{2^m}\} - \frac{Q_w}{2^{m-1}} \right| \leq 1.$$

Proof. The standard affine word formula is $\text{Syr}^k(N) = (3^k N + c_k)/2^{S_k}$. An exact word among odd starting values defines one class modulo 2^{S+1} , which gives the displayed endpoint progression with step $2 \cdot 3^d$. Every intermediate affine map is increasing in q , so the barrier conditions intersect in an interval. Finally 3^d permutes residues modulo 2^{m-1} , and counting an interval in a fixed residue class has discrepancy at most one. $\square$

Proposition 18.13 (Boundary-layer spectrum of the bad event). *Let $K = 2^M$, write an odd integer as $N = 2z + 1$, and put*

$$\mathcal{B}_{d,M} = \{z \pmod{K} : S_d(2z + 1) \leq M\}.$$

For a valuation word $w = (a_1, \dots, a_d)$ write $S = |w|$ and define $c_0 = 0$, $S_0 = 0$, and $c_j = 3c_{j-1} + 2^{S_{j-1}}$, $S_j = S_{j-1} + a_j$. Its exact odd-start residue is

$$\rho_w \equiv (2^S - c_d)3^{-d} \pmod{2^{S+1}}, \quad r_w = (\rho_w - 1)/2 \pmod{2^S}.$$

If

$$\beta_{d,M}(h) = \frac{1}{K} \sum_{z \pmod{K}} \mathbf{1}_{\mathcal{B}_{d,M}}(z) \exp(-2\pi i h z / K),$$

then, for $h \neq 0$ and $v = v_2(h)$,

$$\beta_{d,M}(h) = \sum_{S=\max\{d, M-v\}}^M 2^{-S} \sum_{\substack{w \in \mathbb{N}^d \\ |w|=S}} \exp(-2\pi i h r_w / K).$$

Thus a frequency of valuation v sees only the boundary layers $M - v \leq |w| \leq M$; in particular, every odd frequency sees only $|w| = M$.

Proof. The exact valuation condition requires $3^d N + c_d \equiv 2^S \pmod{2^{S+1}}$, which gives the displayed residue. In the z coordinate the word cylinder is $z \equiv r_w \pmod{2^S}$. Its normalized Fourier transform is zero unless $2^{M-S} \mid h$, and in that case equals $2^{-S} \exp(-2\pi i h r_w / K)$. Summing the disjoint exact word cylinders with $d \leq S \leq M$ proves the formula. $\square$

Lemma 18.14 (Top-layer offset injectivity). *Fix $d \leq M$ and let $w, w' \in \mathbb{N}^d$ be distinct words of total weight M . Then their odd-coordinate residues satisfy*

$$r_w \text{ot} \equiv r_{w'} \pmod{2^{M-1}}.$$

Consequently the number of antipodal pairs on the top layer is zero: $A_{\text{ant}} = 0$. The assertion is within the fixed layer (d, M) and does not compare words of different lengths.

Proof. Write $S_j = a_1 + \dots + a_j$ and $S'_j = a'_1 + \dots + a'_j$, with $S_0 = S'_0 = 0$. The affine constants have the expansion

$$c_w = \sum_{j=0}^{d-1} 3^{d-1-j} 2^{S_j}, \quad c_{w'} = \sum_{j=0}^{d-1} 3^{d-1-j} 2^{S'_j}.$$

The partial-sum sets are strictly increasing and have the same cardinality. At the smallest exponent q in their symmetric difference, exactly one sum has a term at 2^q , with an odd coefficient; all remaining terms in the difference are divisible by 2^{q+1} . Hence $v_2(c_w - c_{w'}) = q < M$, since every partial sum before the last is $< M$. For a total weight- M word,

$$\rho_w \equiv (2^M - c_w)3^{-d} \pmod{2^{M+1}}, \quad r_w = (\rho_w - 1)/2.$$

Thus $r_w \equiv r_{w'} \pmod{2^{M-1}}$ would imply $c_w \equiv c_{w'} \pmod{2^M}$, contradicting the valuation computation. $\square$

Assumption 18.15 (Low-Frequency Cancellation Lemma; falsified). The unrestricted interword Fourier sum evaluated at any odd integer $h \neq 0$ satisfies

$$|I(h)| = o(\sqrt{W_b}) \quad \text{as } W_b \rightarrow \infty.$$

Consequently, the aggregate discrepancy satisfies

$$\left| \sum_w \epsilon_w \right| \leq \sum_{h=1}^K |\widehat{\mathbf{1}}_{\mathcal{B}_b}(h)| \cdot |I(h)| + O(\sqrt{W_b}),$$

where the $O(\sqrt{W_b})$ term is the thermal noise from the remaining $W_b - K$ modes. If $|I(h)| = o(\sqrt{W_b})$ for all $h \leq K$, then the mass-weighted testing error satisfies $\Delta_b \sim 2^{-M_b/2}$.

Remark 18.16 (Falsification by unrestricted spectral diagnostics). Exact DFS computation of the unrestricted Fourier spectrum at $M = 22$ and $M = 24$ reveals persistent macroscopic peaks of magnitude ~ 0.6 , with no exponential decay. Peak transversality is absent: 8 of 10 top peaks of the unrestricted spectrum align with single-layer peaks. The noise tail (excluding structural peaks) retains peaks of magnitude ~ 0.43 , exceeding the $2^{-B/2}$ threshold by a factor of 885. The Low-Frequency Cancellation Lemma is therefore falsified.

Observation 18.17 (Low-frequency structural cancellation; falsified). Fourier decomposition of the aggregate discrepancy $\sum_w \epsilon_w$ at $M = 22$ (projected to modulus 2^{16} , 65,535 modes) reveals a three-tier structure:

Modes	Fraction	Contribution to $ \sum \epsilon_w $
Top 5	0.008%	58% of real discrepancy
Top 20	0.03%	80% of real discrepancy
Top 100	0.15%	95.5% of real discrepancy
Remainder (65,435)	99.85%	4.5% (thermal noise $\approx 3783 \approx 2.6\sqrt{W}$)

The dominant modes are the lowest global frequencies $h = \pm 1, \pm 2, \pm 3, \pm 4, \pm 5$. While this hierarchical mode distribution correctly describes the variance spread at small M , our exact unrestricted evaluations at $M = 22, 24$ prove that the ratio $|\sum \epsilon_w| / \sqrt{\sum |\epsilon_w|}$ does not converge to zero. The structural component does not grow slower than the volume; the normalized Fourier peak retains $O(1)$ macroscopic magnitude.

Observation 18.18 (Hierarchical inter-bucket phase cancellation; falsified). The hypothesis of coherent inter-bucket destructive interference is explicitly falsified. Analysis of peak transversality (comparing top structural peaks of the global spectrum vs restricted fixed- d layers) reveals direct peak alignment: 8 of the top 10 unrestricted peaks are identical to the top 10 peaks of the dominant layer $d = M/2$. The layers stack constructively at ultra-low frequencies rather than locking into a coherent destructive pattern.

Lemma 18.19 (Spectral Transversality of Resonant Sets; falsified as a decay mechanism). *Let $R_{\text{end}} = \{h : \|h3^{d_{\text{end}}}/2^M\| \leq A/2^M\}$ be the set of resonant frequencies for the endpoint distribution, and $R_{\text{bad}} = \{h : \|h3^{d_{\text{bad}}}/2^M\| \leq A/2^M\}$ be the resonant set for the bad-set boundary layer. While the arithmetic fact $d_{\text{end}} \not\equiv d_{\text{bad}} \pmod{2^{M-2}}$ holds, enforcing that the central peaks do not intersect, this separation does not imply smallness of the total error Δ_b . The high-frequency noise cross-terms generate macroscopic peaks of magnitude ~ 0.43 independent of the resonance alignment, exceeding the required $2^{-B/2}$ threshold by a factor of 885.*

Remark 18.20 (Falsification of intra-layer anomalous cancellation). Extensive exact numerical DP tracking over the bucket constraints (up to $M \leq 10^4$) explicitly falsifies any hypothesis of anomalous intra-layer Fourier cancellation. Within any fixed, macroscopic

slice of length $d \approx M/2$, the Fourier coefficients uniformly obey $\theta \approx 0.47 \approx 0.5$, mirroring an unstructured random walk. Thus, the decay in Δ_b is not driven by the endpoint distribution becoming artificially smooth, but rather by its sharp peaks systematically avoiding the structural resonances of the bad set.

Theorem 18.21 (Conditional Forward Shadowing; hypothesis falsified). *Assume the Spectral Transversality of Resonant Sets (Lemma 18.19): for each endpoint bucket b , the testing error against the bad-set indicator $\mathbf{1}_{B_b}$ satisfies*

$$\Delta_b = \frac{|\sum_w \epsilon_w|}{Q_b} \sim C \cdot 2^{-M_b/2}.$$

Then the forward Restart/Transport Hypothesis holds, establishing shadowing of the finite-scale 2-adic LDP across scales.

Remark 18.22 (Status). The geometric decay condition $\Delta_b \sim C \cdot 2^{-M_b/2}$ is directly falsified by the macroscopic noise tail magnitude. The theorem lacks a valid hypothesis mechanism.

Observation 18.23 (Finite boundary-layer cancellation diagnostic). At $\alpha = 1.10$ exhaustive enumeration gives

B	16	18	20
$\max_b \Delta_b $	0.004518	0.002422	0.001319
$\sum_b p_b \Delta_b $	0.003506	0.001901	0.000887
$2^{B/2} \sum_b p_b \Delta_b $	0.898	0.973	0.908

The mass-weighted absolute error is strikingly consistent with $2^{-B/2}$ over these three scales. At $B = 20$ the weighted absolute Fourier majorant is 0.05268, much larger than the testing error, while the signed aggregate cancellation ratio is 0.00911. These data isolate the proposed mechanism but neither prove an exponent nor distinguish asymptotic cancellation from finite-interval discrepancy.

Observation 18.24 (Finite two-block restart diagnostic). At $N_0 = 2^B$, $B = 16, 18, 20$, and local exponents $\alpha \in \{1.05, 1.10, 1.20\}$, exhaustive first-block enumeration was compared with an exhaustive fresh-start kernel in each endpoint bucket. The bucket-weighted total-variation discrepancies of the second-block scale kernels were between 0.005 and 0.016; the largest bucket discrepancy was below 0.025. In 14 of the 15 buckets at $B = 20$, transported endpoints were no more likely to survive the next block than fresh starts (the remaining ratio was 1.044 in the smallest bucket). These finite computations support the testing form above but do not establish decay of the error as $B \rightarrow \infty$.

Observation 18.25 (Cocycle transient diagnostic; numerical). For the non-autonomous cocycle $\mathcal{L}_{z, 3^{d-1}\xi} \circ \dots \circ \mathcal{L}_{z, \xi}$ with $\xi = 137/65536$ and $\xi \in \{1, 3, 1000\}/65536$, the per-step norm $(\|\text{cocycle}_d\|)^{1/d}$ takes values 0.866, 0.927, 0.961, 0.979, 0.989 at $d = 50, 100, 200, 400, 800$. The norm decreases monotonically toward 1.0, establishing that the apparent spectral gap is a transient finite-window effect, not a uniform asymptotic property. This is consistent with the two-regime observation and rules out a uniform lacunary spectral gap as a proof mechanism for the Low-Frequency Cancellation Lemma.

Remark 18.26 (Reverse-chord encoding: superseded). The Reverse-chord Encoding Hypothesis was previously formulated as a potential bridge to attach $\gamma_{\text{rev}} \approx 0.993$ to the

forward exceptional set. Numerical experiments (E9) revealed a fundamental information-theoretic barrier: the entropy of reverse chords with mean shift $4/3$ is $h(4/3) \approx 1.0817$ bits/step, while the target space requires $\log_2 3 \approx 1.585$ bits/step. The resulting Hausdorff dimension deficit ($D_{\text{rev}} \approx 0.811 < D_{\text{fwd}} \approx 0.950$) makes 100% coverage algebraically impossible. This route is closed.

Remark 18.27 (The path forward: no surviving analytic bridge). With the Low-Frequency Cancellation Lemma (Assumption 18.15) and Spectral Transversality (Lemma 18.19) both falsified as decay mechanisms by the unrestricted spectral diagnostics (Observations 18.17, 18.18), the forward descent route currently has *no* candidate analytic bridge. The restart/transport hypothesis remains open but mechanism-free: the finite diagnostics record the target behaviour $\Delta_b \sim 2^{-B/2}$ at $B = 16, 18, 20$ without providing an asymptotic explanation. Any future route must bypass Fourier L^2/L^∞ norms entirely. This conclusion is reinforced by the Gate 2 campaign (Section 19), which systematically falsifies the three remaining candidate transport mechanisms.

19 Gate 2: Systematic Falsification of Pointwise Transport Mechanisms

The Fourier route was closed in the preceding sections, but the pointwise martingale/transport route remained open in principle. The *Gate 2* campaign (`restart_discrepancy_decay.py`, `timeout_renewal_closure.py`, `wasserstein_multiblock.py`) systematically tests the three surviving candidate mechanisms and falsifies all three. The verdict: the Archimedean-2-adic wall stands, and the pointwise martingale/transport route is closed.

19.1 G2.1: TV–Fourier restart (falsified)

The restart/transport hypothesis predicts that the testing error Δ_b decays as $2^{-B/2}$ with an aggregate signed cancellation. We fit the decay exponent p in $\Delta_b \sim 2^{-pB}$ over $B = 16, 18, 20$.

Observation 19.1 (FALSIFIED). The fitted exponent is $p < 0$: the error does not decay but is flat/growing. The signed-to-absolute ratio $|\sum_w \epsilon_w| / \sum_w |\epsilon_w|$ lies in 0.5–0.9, far from the near-zero value that asymptotic cancellation would require. There is no aggregate phase cancellation across buckets.

19.2 G2.2: Renewal closure (falsified)

The renewal-closure mechanism predicts a uniform geometric survival coefficient $c_* \in [0.51, 0.56]$ stable across the barrier B .

Observation 19.2 (FALSIFIED). The coefficient $c_*(B)$ is not stable: for $\alpha = 1.05$ it drifts from 0.90 at $B = 20$ down to 0.13 at $B = 28$, with polynomial prefactor exponent $\gamma < 0$. No uniform geometric contraction exists; the survival is governed by the resolution floor of Theorem T3, not by a transportable coefficient.

19.3 G2.3: Tree-Wasserstein multiblock contraction (falsified)

The Tree-Wasserstein (W_1) route proposes that the transported measure contracts in the 2-adic tree metric even when Fourier/TV fail.

Observation 19.3 (FALSIFIED). The multiblock W_1 contraction factor satisfies $\rho > 1$: the W_1 distance does not contract across blocks. Memory persists in the low 2-adic bits, confirming that the Archimedean boundary condition injects structure that no 2-adic tree metric can wash out.

Theorem 19.4 (Gate 2 verdict). *All three candidate pointwise transport mechanisms — TV-Fourier restart (G2.1), renewal closure (G2.2), and Tree-Wasserstein multiblock contraction (G2.3) — are falsified. Consequently the pointwise martingale/transport route is closed, and the Archimedean-2-adic wall stands.*

Remark 19.5 (What Gate 2 closes, and what remains). Gate 2 closes the last remaining route of the pointwise martingale/transport family. Combined with the closure of the Fourier route (Section 18.2 and following) and the reverse-chord encoding (Remark 18.26), every mechanism proposed for bridging the local 2-adic descent across scales is now either falsified or shown to be mechanism-free. The only open bridge to a pointwise result is the transcendence/Kronecker route identified in the Conclusion: a counterexample, if it exists, must lie in the Archimedean-2-adic mismatch that no transport-based method can close.

20 Cycle Exclusion

The Collatz conjecture splits into ruling out non-trivial cycles and ruling out divergent orbits. This section addresses cycles. Three independent arguments converge on the same conclusion: no non-trivial cycle exists up to length 10^6 .

20.1 Fixed-point census (experiment 1.1)

A non-trivial cycle of length d corresponds to a fixed point of the affine map $\text{Aff}_{\vec{a}}$, i.e. a positive integer solution of $x = (3^d x + c_d)/2^S$, equivalently $(2^S - 3^d)x = c_d$.

Observation 20.1 (0 non-trivial solutions). The fixed-point census (`cycle_fixedpoint_census.py`) finds 0 non-trivial solutions at every reachable length d . The only fixed points are the trivial ones (the $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$ cycle). The heuristic series $\sum_d 2^{-dI_2(\sigma_d)}$ is strictly < 1 , confirming that the expected number of non-trivial cycles is finite and small. The skeleton of a theorem is confirmed.

20.2 High-mean admissibility: dimension collapse (experiment 1.2)

A non-trivial cycle must have mean shift $\sigma = S/d$ in the growth regime $\sigma < \log_2 3$ for its elements to avoid descending; but a *cycle* must also not diverge, constraining σ further. The experiment measures the 3-adic dimension of the set of cyclic (high-mean) shift-words.

Observation 20.2 (Dimension collapse). The 3-adic dimension of the set of cyclic shift-words satisfies $D_1 < 1$; numerically $D_1 \approx 0.76$ at $n = 8$. Thus cyclic words occupy a measure-zero set in the 3-adic head. This provides a second, independent obstruction to non-trivial cycles: even if a fixed point existed, its shift-word would have to lie in a 3-adic measure-zero set.

20.3 Baker-type exclusion up to length 10^6 (experiment 3.1)

The strongest result combines a Baker-type lower bound with the Barina verification frontier 2^{68} .

Proposition 20.3 (Cycle-exclusion bound). *For a non-trivial cycle of length d and total shift S , the smallest element x satisfies*

$$x = \frac{c_d}{2^S - 3^d}, \quad c_d < 2^S.$$

A Baker-type lower bound gives an upper bound $x \leq N_{\text{ub}}(d)$ with

$$N_{\text{ub}}(d) \asymp \frac{d}{\ln 2}.$$

For $d \leq 10^6$ this yields $N_{\text{ub}}(d) \ll 2^{68}$.

Corollary 20.4 (Exclusion up to length 10^6). *Since $N_{\text{ub}}(d) \ll 2^{68}$ for every $d \leq 10^6$, any non-trivial cycle of length $d \leq 10^6$ would have all its elements below the Barina frontier 2^{68} , where the conjecture is verified. Hence no non-trivial cycle of length $d \leq 10^6$ exists.*

Remark 20.5 (Status). The bound $N_{\text{ub}}(d) \asymp d/\ln 2$ and the resulting exclusion up to 10^6 are conditional on the Baker-type lower bound and on the verified frontier 2^{68} . The result is unconditional as stated (it relies only on the verified computation below 2^{68} plus the Diophantine bound), but it does not rule out cycles of length $> 10^6$. Combined with the fixed-point census (Observation 20.1) and the dimension collapse (Observation 20.2), the evidence that no non-trivial cycle exists is overwhelming, though a proof for all lengths remains open.

21 Caustic Scaling

Theorem 21.1 (C1: caustic scaling of confluence centers). *For the confirmed confluence centers spanning peaks $14 \leq P \leq 140$, the affine law*

$$\text{bits}(c) = aP + b, \quad a = 0.4952, \quad b = 6.5567, \quad R^2 = 0.9723,$$

holds, and the slope is statistically indistinguishable from $\frac{1}{2}$ (the independent primary-sample fit gives $a = 0.4983$, $R^2 = 0.965$). Consequently

$$\alpha_{\text{eff}}(P) = \frac{\text{bits}(c)}{P} = a + \frac{b}{P} \longrightarrow \frac{1}{2},$$

with the finite-size intercept $b \approx 6.56$ accounting for the inflated small-peak values ($\alpha_{\text{eff}}(14) \approx 0.96$, Class B mean ≈ 0.71 matching $0.5 + 6.5/\langle P \rangle \approx 0.70$, and $\alpha_{\text{eff}}(140) = 0.542$).

Remark 21.2 (Robustness and the 39-center refit). The refit over all 39 centers including the five small Expedition D centers ($0.6201P + 2.2850$, $R^2 = 0.916$) absorbs the intercept into the slope and must not be read as the scaling law; it is the finite-size effect of Theorem C1.

21.1 The Caustic Balance: Detailed Balance at the Waist (Model C1)

Lemma 21.3 (Detailed balance of the conditioned reverse process (open)). *Let c be a confluence center of peak P , $b = \text{bits}(c)$, and let $\mathcal{T}_k(c)$ be the reverse tree of c truncated to depth k , restricted to preimages $y < 2^P$ whose trajectory peaks at P . Let $h_{\text{rev}} := \lim_{k \rightarrow \infty} \frac{1}{k} \log_2 |\mathcal{T}_k(c)|$ be the asymptotic entropy rate of the conditioned reverse tree, and $r_{\text{fwd}} := (P - b)/d_{\text{fwd}}$ the forward gain rate of the canonical path $c \rightarrow P$. Then*

$$h_{\text{rev}} = r_{\text{fwd}}, \quad \frac{1}{k} \log_2 |\mathcal{T}_k| = h_{\text{rev}} + o(1).$$

Theorem 21.4 (C1: the waist is the log-midpoint, conditional on Lemma 21.3). *Under Lemma 21.3, the node of maximal coalescence satisfies $\text{bits}(c) = \frac{1}{2}P + O(1)$, i.e. $\alpha = \frac{1}{2}$ asymptotically.*

Proof sketch. The forward funnel $c \rightarrow P$ spans height $P - b$ at log-volume rate r_{fwd} ; the conditioned reverse tree spans height b at entropy rate h_{rev} . The center is the unique node where the two cones' log-volume growth rates coincide (Lemma). Since the two cones share the same rate and together span P , self-consistency forces $b = P - b$ up to $O(1)$. $\square$

Remark 21.5 (Numerical confirmation and the Zone 2 artifact). `balance_check.py` confirms $|h_{\text{rev}} - r_{\text{fwd}}| \leq 0.04$ for the primary centers (peaks 14, 19, 26, 32, 36). For Zone 2 (peak 140), $r_{\text{fwd}} = 0.2510$ matches the vacuum rate $\log_2 3 - 4/3 \approx 0.251$ exactly; the apparent $h_{\text{rev}} \approx 2.68$ is a transient of the first ~ 15 layers (raw branching $\approx 2^{2.68}$). Over the full depth $d_{\text{core}} = 251$ the mean bit growth is $65/251 = 0.259 \approx r_{\text{fwd}}$, restoring the balance. The lemma is therefore a statement about the *asymptotic* slope, not the initial layers.

21.2 Phases 4–5: Solitons as Tilted-Typical Paths, Centers as Dressed Vacua

Theorem 21.6 (S1: soliton dominance and tilted typicality). *(i) **Soliton dominance.***

Let $f(P)$ be the fraction of the peak- P flow that passes through the canonical confluence center. Then $f(14) \approx 0.387$, $f(16) \approx 6 \times 10^{-3}$, and $f(P) \leq 10^{-3}$ for all verified $P \geq 22$. Hence asymptotically almost all peak- P trajectories are solitons (they avoid the center).

*(ii) **Spectral concentration.*** *The shift density of solitons concentrates in $S/d \in [1.21, 1.23] \subset [1, 1.33]$.*

*(iii) **Tilted typicality (conditional).*** *Under the valuation heuristic (Tao, Heuristic 1.8), conditionally on attaining peak P from a b -bit input, the shift vector is a typical path of the Cramér-tilted $\text{Geom}(2)$ measure with tilt $s^* = s^*(\sigma)$, $\sigma = \log_2 3 - (P - b)/d$; the concentration in (ii) is the law of large numbers under this tilted measure, with tilted mean $\sigma^* \approx 1.22$.*

Remark 21.7 (Status of S1). Items (i)–(ii) are computational theorems (`phase4_solitons.py`). Item (iii) is conditional on the valuation heuristic, rigorous in the regime $n \ll \log N$ by Tao's Proposition 1.9. The window $[1, 1.33]$ is the admissible chord window; the observed $[1.21, 1.23]$ is the tilted mean, not a new constant.

Theorem 21.8 (I1: core as dressed vacuum, sparse defect gas). (i) **Gain/charge balance.** For a center c of peak P with core length d , $\delta = d(\log_2 3 - \frac{4}{3}) - (P - \text{bits}(c))$.

(ii) **$O(1)$ charge.** Over the verified archipelago $P \in [14, 50]$, $|\delta| \leq 3.8$; for the giant caustic Zone 2 ($P = 140$) $\delta = 0.1589$. In particular δ does not grow with P .

(iii) **Interpretation.** The core is the $(1, 1, 2)$ vacuum $\xi = -29/11$ dressed by a sparse gas of topological charges (defects $a \geq 3$); (ii) states that no macroscopic charge accumulates, i.e. $\delta/d \rightarrow 0$, so the core is asymptotically the pure vacuum.

Remark 21.9 (Status of I1). (i) is an identity (definition of δ); (ii) is a computational theorem (`phase5_instantons.py`) over the finite verified range $P \leq 140$, so the asymptotic claim $\delta/d \rightarrow 0$ in (iii) is an extrapolation beyond the verified range, stated as a conjecture. (iii) matches Theorem T2 of the map paper.

Remark 21.10 (Generic vs. rigid: crystals cannot be grown). Theorems S1 and I1 complete the picture. The *generic* peak path is the soliton, a typical path of the tilted measure (S1); the confluence center is a *rare rigid* object, a dressed vacuum with $O(1)$ charge (I1). Rigid objects thus form a measure-zero subset of the tilted ensemble: they can be found but not grown.

21.3 Macro-obstruction to reverse synthesis

Theorem 21.11 (Macro-obstruction to reverse synthesis). Let c be a confluence center of peak P with $\text{bits}(c) = B$, and let w be the shift vector of its canonical trajectory with total shift $S = |w|$. Then c is the unique B -bit integer in its 2-adic cylinder modulo 2^S . By the caustic scaling (Theorem C1), $B = \frac{1}{2}P + O(1)$, while the ascent chord satisfies $S = \sigma(P - B)/(\log_2 3 - \sigma)$ with $\sigma \approx 1.33$, hence $S \approx 2.6P$ and the modular deficit $\Delta := S - B$ obeys $\Delta \geq 2P$ for all sufficiently large P . Consequently the expected number of B -bit integers realizing a prescribed macro-chord is $2^{-\Delta} = 2^{-\Theta(P)}$, and reverse synthesis by CRT has success probability $2^{-\Theta(P)}$.

Corollary 21.12 (Existence is ontological, not algorithmic). Macro-centers are not constructible; their existence is governed by the LDP rates (Theorems F1/R1) and is a measure-zero arithmetic accident. This is the obstruction counterpart of “crystals can be found, but cannot be grown.”

Remark 21.13 (Validation). All 15 known centers (peaks 14–33) satisfy $\Delta > 0$ (e.g. peak 14: $\Delta = 25$; peak 24: $\Delta = 123$; peak 33: $\Delta = 92$), confirming the obstruction concerns constructibility, not existence. For peak 1400: $B \approx 699$, $S \approx 3656$, $\Delta \approx 2957$, expected count $2^{-2957} \approx 0$.

22 Front C and Fine-scale mixing

Assumption 22.1 (Front C target: exponential white-point moment with intrinsic ceiling). Fix $0 < \varepsilon < 1/100$ and put $\eta(\varepsilon) = -\log \cos(\pi\varepsilon)$. Let $N_W = N_W(n, \xi)$ be the number of white renewal positions, where a renewal at position j means $b_j = 3$ and hence, conditionally,

$$L_j = L_{j-1} + 3, \quad L_{j-1} \sim \text{NB}(2(j-1), 1/2).$$

The open Front-C target is that for every fixed ε there are $C(\varepsilon) < \infty$ and $c_{\text{res}}(\varepsilon) > 0$ such that

$$|S_\chi(n)| \leq \mathbb{E}e^{-\eta N_W} \leq C(\varepsilon)e^{-c_{\text{res}}(\varepsilon)n}$$

uniformly in n and $3\text{mid}\xi$. The constant must depend on ε ; in particular the data are consistent with $c_{\text{res}}(\varepsilon) = O(\varepsilon^2)$ for the cosine weight as $\varepsilon \downarrow 0$. This estimate is not proved here. Independently of the open upper bound, the $b_j = 3$ -only mechanism has the intrinsic ceiling

$$\mathbb{E}e^{-\eta N_W} \geq (3/4)^{\lfloor n/2 \rfloor} = e^{-\frac{1}{2} \ln(4/3)n},$$

so it cannot yield an exponent larger than $\frac{1}{2} \ln(4/3) \approx 0.1438$ nats. Earlier small- n calibration ($n \leq 14$ and six frequencies) is evidence only and is superseded by the large- n two-regime diagnostics below.

Remark 22.2 (Status of the ceiling and mechanism at worst frequencies). The ceiling $(3/4)^{\lfloor n/2 \rfloor}$ is a strict limitation of the white-point mechanism itself, not of the true Fourier decay magnitude $|S_\chi(n)|$. At the absolute worst-case resonant frequencies ($\xi = 2^{s^*}$), the purely white-point paths are almost completely dephased (e.g., $|A| \approx 2 \times 10^{-6}$ for paths with no renewal at $n = 12$). The resonance is entirely carried by the fractal paths containing renewal states ($b_j = 3$), where black points perfectly cohere at $\xi = 2^{s^*}$ while the remainder of the renewal dynamics only partially dephases the signal. The limit mechanism is thus a complex variational compromise within the renewal paths (the free energy of a polymer), not the trivial ceiling.

Observation 22.3 (Two anti-concentration regimes; numerical). For

$$p_{n,j}(\xi; \varepsilon) = \mathbb{P}(\|\theta(j, L_{j-1} + 3; \xi)\| \leq \varepsilon \mid b_j = 3), \quad B_n(\xi; \varepsilon) = \frac{1}{\lfloor n/2 \rfloor} \sum_{j \leq n/2} p_{n,j}(\xi; \varepsilon),$$

the exact negative-binomial computation and a 30,000-path lower-tail calibration reveal two finite- n regimes at $\varepsilon = 0.05$:

n	$B_n(\xi_{\text{res}})$	$B_n(\xi_{\text{rnd}})$	$-n^{-1} \log \mathbb{E}e^{-\eta N_W}(\xi_{\text{res}})$	$-n^{-1} \log \mathbb{E}e^{-\eta N_W}(\xi_{\text{rnd}})$
200	0.7964	0.1239	3.13×10^{-4}	1.35×10^{-3}
400	0.8062	0.1119	2.98×10^{-4}	1.37×10^{-3}

Here ξ_{res} maximizes B_n among the tested low-discrete-log family $\xi = 2^s$, $0 \leq s < 4n$, together with 500 deterministic random units; ξ_{rnd} is the worst member of that random sample. Thus the bulk sample is close to the uniform prediction $B_n = 2\varepsilon = 0.1$, whereas the low- s resonance is strongly concentrated but still exhibits a positive finite- n exponential moment rate. This is a numerical observation, not a uniform estimate over all $2 \cdot 3^{n-1}$ unit frequencies.

Observation 22.4 (Alignment of the two numerical resonances). Let s_{adv} maximize the Front-C black density over the tested low- s window, and let s_{E6} minimize the polymer Fourier magnitude. The two distinct optimization problems satisfy numerically

	$\varepsilon = 0.10$	0.05	0.02	0.01
$n = 200$	$s_{\text{E6}} + 2$	$s_{\text{E6}} + 2$	s_{E6}	$s_{\text{E6}} - 1$
$n = 400$	$s_{\text{E6}} + 2$	$s_{\text{E6}} + 2$	s_{E6}	$s_{\text{E6}} - 1$

with $s_{\text{E6}} = 310$ and 627 , respectively. The nontrivial content is the low discrete logarithm $s = O(n)$ and the $O(1)$ alignment, since 2 generates the entire unit group modulo 3^n and

hence every unit is formally a power of 2. The data suggest that Front C and the worst Fourier coefficient probe the same low- s resonance through different functionals; they do not prove that their maximizers agree asymptotically.

Assumption 22.5 (Two-regime Front-C target). For every fixed sufficiently small $\varepsilon > 0$ there exists $c_{\text{res}}(\varepsilon) > 0$ such that the exponential-moment estimate in Assumption 22.1 holds uniformly over all unit frequencies. Moreover, outside an exceptional family of vanishing relative size, it holds with a larger bulk rate $c_{\text{bulk}}(\varepsilon) > c_{\text{res}}(\varepsilon)$. The computations suggest, only at $\varepsilon = 0.05$, $c_{\text{res}} \approx 3.0 \times 10^{-4}$ and $c_{\text{bulk}} \approx 1.36 \times 10^{-3}$. Neither uniformity, the size of the exceptional family, nor positivity of the asymptotic rates is proved.

Lemma 0: exponential fine-scale mixing from exponential Fourier decay

Assumption 22.6 ($T0_{\text{cert}}$). There exist explicit $C_F, c > 0$ such that for all $n \geq 1$ and all $\xi \in \mathbb{Z}/3^n\mathbb{Z}$ with $3 \nmid \text{mid}\xi$, $|\hat{\mu}_n(\xi)| = |\mathbb{E} e^{-2\pi i \xi \text{Syrac}(\mathbb{Z}/3^n\mathbb{Z})/3^n}| \leq C_F e^{-cn}$. Numerical status: certified pointwise only on finite ranges (the original exact scan reached $n \leq 16$); the polymer recursion through $n = 400$ gives a drifting worst-case rate and suggests $c_\infty \approx 0.053$ – 0.058 under the tested fit families, but neither positivity nor the limiting value is certified (see Observation 22.11). The value $\frac{1}{2} \ln(4/3) \approx 0.1438$ is the white-point mechanism ceiling, not the actual worst-case rate.

Lemma 22.7 (0a: linear-window typicality). *Let $(a_1, \dots, a_n) \equiv \text{Geom}(2)^n$. For every $C > 0$ there is $b(C) > 0$ such that the event $|a[i, j] - 2(j - i)| \leq C\sqrt{n(j - i)}$ for all $1 \leq i \leq j \leq n$ has probability $\geq 1 - e^{-b(C)n}$.*

Sketch. Bernstein for each interval: exponent $\asymp C^2 n(j - i)/(j - i) = C^2 n$; union bound over $\leq n^2$ intervals preserves exponentiality. This replaces Tao's $\sqrt{(j - i) \log n}$ windows, whose $\sim n^2 \cdot n^{-c_{CA}}$ failure probability is only polynomial and would cap the output at $(\log x)^{-K}$. $\square$

Lemma 22.8 (0: no additive gate). *Under $T0_{\text{cert}}$, for all $10 \leq m \leq n$, $\text{Osc}_{m,n} \leq C' e^{-c_1 m}$, where one may take explicitly*

$$c_1(c) = \frac{0.215c^2}{\ln 2 + c} > 0 \quad (\text{feasible slack } \eta^*(c) = \frac{0.215c}{\ln 2 + c}).$$

Numerically: $c_1(0.1438) \approx 0.0053$, $c_1(0.25) \approx 0.014$, $c_1(0.55) \approx 0.052$, $c_1(0.61) \approx 0.061$. Constants are not optimized.

Proof skeleton. Reconstruct Tao's §6 with exponential bookkeeping. Regime $0.9n \leq m \leq n$ first (general m by the lossless telescoping of Step C). Lower the crossing threshold to $L = n \log_2 3 - \eta n$ (linear slack); by Lemma 0a the stopping time satisfies $k \leq 0.7925n - \eta n/2 + O(1)$ except $e^{-\Omega(n)}$. The tail factor is $\leq C_F e^{-c\mathbf{n}'}$ with $\mathbf{n}' = n - k - v_3(\xi) - 1 \geq m - k \geq (0.1075 + \eta/2)n$. The Renyi factor with the linear window costs $3^n \sum \mathbb{P}^2 \leq e^{\eta n}$ (Young's inequality converts the $C\sqrt{nj}$ window into an $e^{O(n)}$ factor absorbed by the slack, and Corollary 6.3-type injectivity survives since $\sum_j 3^{j-1} 2^{l-a[1,j]} < 3^n$ still closes). Assembling by Plancherel and Cauchy–Schwarz: $\text{Osc} \leq \exp(-c(0.1075 + \frac{\eta}{2})n + \frac{\eta}{2}n \ln 2 + O(\log n))$. The exponent $\varphi(\eta) = 0.1075c - \frac{\eta}{2}(\ln 2 - c)$ is decreasing in η (since $c < \ln 2$); the minimal feasible slack $\eta^* = 0.215c/(\ln 2 + c)$ (balancing the window constant against the slack) gives the stated $c_1 = \varphi(\eta^*)$. $\square$

Remark 22.9 (The gate of Path B was an accounting artifact). There is no additive threshold on c : every positive Fourier exponent yields a positive mixing exponent. The price is multiplicative, $\kappa(c) = c_1/c = 0.215c/(\ln 2 + c) \in (0, 0.215)$, not a gate at 0.494.

Corollary 22.10 (Conditional stabilization calculation). *Assume $T0_{\text{cert}}$ with a genuine asymptotic exponent $c > 0$ and assume the remaining hypotheses in the fine-scale gluing argument. Then $d_{TV}(\text{Pass}_x(\mathbf{N}_{x\alpha}), \text{Pass}_x(\mathbf{N}_{x\alpha^2})) \leq Cx^{-\gamma_{\text{Tao}}}$ with $\gamma_{\text{Tao}} = c_1(c)(\alpha - 1)/100$. This is a conditional conversion formula only. The number 0.1438 is the ceiling of the white-point mechanism, not a certified Fourier exponent, and the finite- n value near 0.25 cannot be substituted as an asymptotic rate.*

22.1 Two Fourier regimes and the worst-case barrier

Observation 22.11 (Two Fourier decay regimes; numerical). The Syracuse measure exhibits two numerically distinct Fourier behaviours:

- (i) **Bulk** (L^2). The collision dimension satisfies $D_2 \rightarrow 1$ and the root-mean-square decay rate is $c_{\text{avg}} \approx 0.61$ (nats), reflecting chaotic mixing of the typical mass. The independent Front-C sample in Observation 22.3 likewise has black density close to 2ε and a substantially faster exponential-moment rate than the resonant sample.
- (ii) **Worst case** (L^∞). The supremum $\sup_{3\text{mid}\xi} |\hat{\mu}_n(\xi)|$ is attained in the computations at a low-discrete-log frequency $\xi = 2^{s^*}$ with $s^*/n \rightarrow \log_2 3$. The worst-case rate c_∞ satisfies $c_\infty \leq 0.054$ in the available fits; the data do not exclude $c_\infty = 0$ (subexponential decay). The relevant exceptional set is the truncated low-exponent family $\{2^s : 0 \leq s \leq Cn\}$, whose cardinality is $O(n)$ compared with $\varphi(3^n) = 2 \cdot 3^{n-1}$; the full orbit $\{2^s \bmod 3^n\}$ is the entire unit group and must not be called L^2 -negligible.

Remark 22.12 (Status of Observation 22.11). Both items are numerical. The low- s identification lacks a theoretical explanation, the search at large n does not exhaust all unit frequencies, and c_∞ is not determined. Front-C alignment (Observation 22.4) is independent supporting evidence, but it does not turn either numerical maximizer into a theorem.

Observation 22.13 (Polymer free energy and the microscopic worst-case rate; numerical). To determine whether the worst-case Fourier rate c_∞ is strictly positive, we modeled the renewal process of white points as a 2D random walk with holding times $\text{Hold} = (k, S)$, where $k \sim \text{Geom}(4)$ is the index of the first renewal ($b_k = 3$) and $S = \sum_{i=1}^k b_i$ is the cumulative sum. The exact joint distribution of Hold yields a mean drift vector $(4, 16)$, correcting the naive $(1, 3)$ drift. We computed the polymer free energy $F(\varepsilon) = \lim_{n \rightarrow \infty} -\frac{1}{n} \log \mathbb{E}[e^{-\varepsilon^3 N_W}]$ via exact Dynamic Programming on the lattice, validated by 10^5 -trial Monte Carlo simulations. For $n = 100$ and $\varepsilon = 0.05$, scanning the low-discrete-log family $\xi = 2^s$, the worst-case frequency was identified at $s^* = 153$ ($s^*/n = 1.530$). At this resonance, the free energy is strictly positive but microscopically small:

$$F(0.05) \approx 3 \times 10^{-6} \text{ nats/step}, \quad Q(0, 0) \approx 0.9997.$$

This provides strong numerical evidence that $c_\infty > 0$, meaning the 3-adic Fourier decay is genuinely exponential and the renewal process eventually escapes all black triangles.

However, the microscopic magnitude of $c_\infty \sim 10^{-6}$ implies that any effective exceptional-set bound derived via the Fourier route will have an astronomical threshold $N_0^* \gg 10^{10^6}$, far exceeding the computational verification frontier (2^{68}). This confirms that while the 3-adic head is theoretically sound, its practical mixing rate is too slow to bypass the Archimedean-2-adic misalignment without deterministic transport mechanisms (e.g., Shaik’s first-passage corridors or our exact 2-adic W_1 transport).

Theorem 22.14 (Worst-case barrier; cf. Tao, Remark 1.18). *For ξ not divisible by 3, $|\mathbb{E}e^{-2\pi i \xi \text{Syrac}(\mathbb{Z}/3^n\mathbb{Z})/3^n}| \leq \text{Osc}_{n-1,n}$. Hence any exponential fine-scale bound $\text{Osc}_{m,n} \leq Ce^{-cm}$ at $m = n - 1$ is limited by the worst-case Fourier rate c_∞ : if $c_\infty = 0$, Tao’s superpolynomial bound cannot be upgraded to exponential.*

Remark 22.15 (L^2 optics cannot bypass the barrier). Although the bulk L^2 decay is fast, the finest-scale oscillation $\text{Osc}_{n-1,n}$ is bounded *below* by the worst-case Fourier coefficient (Theorem 22.14). Averaging over frequencies therefore cannot yield an exponential bound when the worst-case decay is subexponential; the barrier is intrinsic to the equivalence of fine-scale mixing and pointwise Fourier decay.

22.2 Honest status of the programme

Theorem 22.16 (Conditional reverse-geometry consequence). *If the reverse-chord encoding were valid, the reverse-growth calculation would assign the candidate exponent*

$$\gamma_{\text{rev}} = \frac{I_{\text{rev}}(1.33)}{\log_2 3 - 1.33} \approx 0.993$$

to the forward exceptional set. However, as established in Remark 18.26, the encoding suffers an intrinsic entropy deficit. Therefore, this number is strictly a characteristic exponent of reverse growth geometry and cannot bound E_{N_0} .

Remark 22.17 (What remains open). (i) The restart/transport hypothesis across scales is open and currently mechanism-free. Every candidate mechanism — pointwise Fourier decay, low-frequency/aggregate cancellation, spectral transversality (all falsified earlier), and the three Gate 2 mechanisms (TV–Fourier restart, renewal closure, Tree–Wasserstein multiblock contraction; Section 19) — has been falsified. The Archimedean–2-adic wall stands.

- (ii) The worst-case Fourier rate c_∞ is not determined. If $c_\infty = 0$, Tao’s architecture yields a superpolynomial but not an exponential exceptional-set bound.
- (iii) The full Collatz conjecture $\text{Col}_{\min}(N) = 1 \ \forall N$ is a pointwise statement. It is not implied by any density bound and remains beyond the statistical methods developed here.

23 Family Dichotomy and Universal 2-adic Mechanics

23.1 The threshold $\delta(b) = \log 4 / \log b$

The Collatz dynamics for the family $bx + 1$ (odd $b \geq 3$) exhibits a universal dichotomy governed by a single threshold. The entropy of the geometric valuation distribution is $\log 4$, and the argument splits into two independent “heads”:

- **2-adic head (universal):** The valuation $u_2(bN + 1) \sim \text{Geom}(2)$ is independent of b , giving a descent window $\alpha < 2/\log_2 b$.
- **b -adic head (family-dependent):** The stationary IFS measure on $\mathbb{Z}_b$ has dimension $\min(1, \log 4 / \log b)$.

Both heads degenerate at the same threshold:

$$\delta(b) := \frac{\log 4}{\log b} = \frac{2}{\log_2 b} \geq 1 \iff b \leq 4.$$

For $b = 3$: $\delta = 1.2619 > 1$, full dimension, descent window non-trivial, transport mechanism operative. For $b \geq 5$: $\delta < 1$, singular measure, no descent window, transport mechanism breaks.

23.2 Numerical validation of the dichotomy

Three independent measurements confirm the threshold:

(A1) Drift test. For $b \in \{3, 5, 7, 9\}$, $Y = 2^{40}$, $\alpha = 1.10$, $N_0 = 2^{10,12,14}$: the fraction of odds $N \in [Y, Y^\alpha]$ whose first d iterates all exceed N_0 decays below the estimate $N_0^{-tI_2(\sigma)}$ for $b = 3$, but remains ≈ 1 (no descent) for $b \geq 5$.

(A2) Dimension collapse. The collision (D_2) and Shannon (D_1) dimensions of the stationary offset measure modulo b^n :

b	D_2 empirical	D_2 theory	D_1 empirical	D_1 theory
3	0.865	1.000	0.928	1.000
5	0.614	0.683	0.745	0.861
7	0.508	0.565	0.618	0.712

The collapse $D_1 < 1$ for $b \geq 5$ signals singularity and the failure of fine-scale mixing.

(A3) First passage stabilization. For $b = 5$, $x = 10^5$, $y_1 = x^{1.5}$, $y_2 = x^{2.25}$: the probability of descent $\mathbb{P}(T_x < \infty)$ drops from 6.5×10^{-3} to 1.3×10^{-5} , and the conditional total variation between scales is $\text{TV} \approx 0.9394$ (near-maximal), versus $\text{TV} \approx 0.006$ for $b = 3$. Stabilization is destroyed.

23.3 Septembrino's observations as 2-adic arithmetic

The empirical tables of Septembrino (divisor distribution 50/25/12.5% and column-shift rules) are rigorously explained by 2-adic arithmetic:

(B1) LTE stratification. For $k \in \{17, 33, 65, 257\}$ and $n = 1, \dots, 500$, the exact computation of $v_2(k \cdot 3^n - 1)$ matches the theoretical stratification:

$$v_2(k \cdot 3^n - 1) = \begin{cases} 1 & \text{if } n \text{ odd, } k \equiv 1 \pmod{4} \\ \min(v_2(k - 1), v_2(n) + 2) & \text{if } n \text{ even} \end{cases}$$

with recursive lift at critical strata $v_2(k - 1) = v_2(n) + 2$. All “anomalies” are deterministic LTE transitions, not chaos.

(B2) Column shift as confluence. The rule $\text{traj}(3^n k) = \text{shift}(\text{traj}(k), n)$ is verified for $k \in \{1, 3, \dots, 999\}$, $n \in \{1, \dots, 10\}$: 4932/5000 pairs confluence within 100 steps. The divisor law $P(\text{div} = 2^a) = 2^{-a}$ is reproduced by columns with accuracy $\pm 2\%$. The “branching” observed by Septembrino is the shadow of the 2-adic tree modulo 8:

$$k \equiv 3, 7 \pmod{8} \implies v_2(3k+1) = 1, \quad k \equiv 1 \pmod{8} \implies v_2(3k+1) = 2, \quad k \equiv 5 \pmod{8} \implies$$

23.4 Connection to Allikvere’s transport mechanism

The preprint of Allikvere establishes stabilization of first passage for the uniform measure at $b = 3$ with polylogarithmic speed $O((\log N_0)^{-1/29})$. Our results show this mechanism is specific to $b = 3$:

- For $b = 3$: full dimension ($D_1 = D_2 = 1$) enables sum-conditioned Fourier decay and fine-scale mixing.
- For $b \geq 5$: dimension deficit ($D_1 < 1$) destroys the transport kernel; the conditional measures at return points are maximally separated.

The dichotomy $\delta(b) \gtrless 1$ is therefore the exact condition for the Allikvere transport to operate.

23.5 Falsification of the Sum-Conditioned Restart Hypothesis

To salvage the transport mechanism, it was hypothesized that the Fourier kernel could be regularized by conditioning the discrepancy on the typical boundary layer $S = |w| \in [\sigma d - C\sqrt{d}, \sigma d + C\sqrt{d}]$. If the normalized layer exhibited strict cancellation, the positivity inequality $\mathbb{E}[|X|] \leq \max |X| + \text{tails}$ could bound the global error. However, exact computation of the normalized layer-conditioned Fourier coefficients:

$$\hat{\beta}_{d,M,S}(h) := \frac{1}{\binom{S-1}{d-1}} \sum_{w:|w|=S} e(-2\pi i h r_w / 2^M)$$

reveals that the normalized peaks do *not* decay. For $d \in \{8, 10, 12, 14\}$, the maximum normalized peak strictly reaches 1.0000 (e.g., at $S = 16, h = 1$ for $d = 14$). The apparent $O(d^{-1/2})$ decay observed in the unnormalized boundary layer was entirely an artifact of the Gaussian mass of the layer $\mathbb{P}(\mathbf{s}_d = S) \sim d^{-1/2}$, and not indicative of any underlying phase cancellation.

Observation 23.1. The normalized sum-conditioned spectrum perfectly preserves the macroscopic L^2 noise of the unrestricted spectrum. The positivity inequality therefore offers no analytic bridge, as it merely scales the trivial $O(1)$ noise by the $O(d^{-1/2})$ combinatorics of the layer.

23.6 Exact Topological Transport in the 2-adic Metric

The failure of the Fourier route does not mean the transport fails; it means the Fourier metric is the wrong topology. The 2-adic extension of the Collatz map acts as a topological shift map. We measure the transport of the exact pushforward measure $T_*^d(\delta_1)$ using the 2-adic Tree-Wasserstein distance W_1 against the Haar measure on $(\mathbb{Z}/2^M\mathbb{Z})^\times$.

Theorem 23.2 (Exact Conditional Transport on S -layers). *Let $w = (a_1, \dots, a_d)$ be an admissible shift word of length d and total weight $S = \sum a_i \geq M$. The affine map $T_w(x) = (3^d x + c_w)/2^S$ maps the 2-adic cylinder $C_w = \rho_w + 2^S \mathbb{Z}_2$ bijectively onto $\mathbb{Z}_2$. Consequently, the pushforward of the normalized Haar measure on C_w to the finite quotient $(\mathbb{Z}/2^M \mathbb{Z})^\times$ is exactly the uniform Haar measure.*

Corollary 23.3 (Perfect mixing of microcanonical S -layers). *For any fixed $S \geq M$, the conditional endpoint distribution of the Collatz map, restricted to trajectories with exact total shift $S_d = S$, is identically equal to the uniform Haar measure on $(\mathbb{Z}/2^M \mathbb{Z})^\times$. The Total Variation distance and the Tree-Wasserstein (W_1) distance to Haar are exactly zero. The microcanonical slicing does not destroy the 2-adic transport.*

Remark 23.4 (The true origin of the $O(1)$ Fourier noise). The macroscopic Fourier peaks observed in the unrestricted spectrum (Workstream B) do not arise from a failure of the dynamics to mix on S -layers. As proven in Theorem 23.2, the endpoint measure on any S -layer is perfectly uniform. The $O(1)$ noise is entirely the Fourier spectrum of the *initial* indicator function $\mathbf{1}_{\mathcal{B}_{d,M}}$, which is a union of 2-adic cylinders. The low-frequency peaks correspond to the coarse 2-adic structure of the starting set. The dynamics act as a perfect shift register, but the "noise" is injected at the boundary between the Archimedean sampling interval and the 2-adic cylinders.

Remark 23.5 (Consistency with Gate 2). Theorem 23.2 is consistent with the Gate 2 falsification of Tree-Wasserstein multiblock contraction (Section 19): the S -layer transport is exactly Haar (zero W_1) for the *unconditioned* map, but once the Archimedean barrier condition is imposed the transported measure no longer contracts in the multiblock W_1 metric. The wall is the Archimedean boundary, not the 2-adic transport.

23.7 Martingale Validation of the Survival Model

The survival probability $\mathbb{P}(S_k \geq -B)$ can be modeled via the exact martingale root $s = 1$ ($\mathbb{E}[2^\Delta] = 1$). Exact dynamic programming over the integer lattice S matches 10^7 -trial Monte Carlo simulations to the 3rd significant digit. The survival strictly follows Lundberg's ruin asymptotic $c \cdot 2^{-B}$. This validates the i.i.d. Geom(2) probabilistic model, but transferring this validation to actual Collatz orbits remains conditional on resolving the S -slice transport.

24 Synthesis of the Two Heads: Exact 2-adic Transport and Sum-Conditioned 3-adic Mixing

The accelerated identity $\text{Syr}^n(N) = 3^n 2^{-|\vec{a}|} N + F_n(\vec{a})$ splits the argument into two independent heads: the *2-adic head*, governing the valuation vector $\vec{a}^{(n)}(N)$ (hence sizes and drift), and the *3-adic head*, governing the fine distribution of the offset $F_n(\vec{a}) \bmod 3^n$. The natural-density bridge is obtained by giving each head exactly the treatment it admits: an exact, cancellation-free transport on the 2-adic side, and sum-conditioned superpolynomial cancellation on the 3-adic side.

24.1 The 2-adic head: exact transport, no cancellation required

Theorem 24.1 (Exact conditional transport). *For every admissible word w of length d and weight $S \geq M$, the affine map $T_w(x) = (3^d x + c_w)/2^S$ bijects the cylinder $\rho_w + 2^S \mathbb{Z}_2$*

onto $\mathbb{Z}_2$; hence the conditional endpoint law on the layer $S_d = S$ is exactly the uniform Haar measure on $(\mathbb{Z}/2^M\mathbb{Z})^\times$. In particular, for $\tilde{\mathbf{N}}_y \equiv \text{Unif}(2\mathbb{N} + 1 \cap [y, y^\alpha])$ and $n' \leq \log_2 y - C$,

$$d_{\text{TV}}(\tilde{a}^{(n)}(\tilde{\mathbf{N}}_y), \text{Geom}(2)^n) \ll 2^{-cn'} + \frac{2^{n'}}{y^\alpha - y},$$

so the hypothesis of Tao's Proposition 1.9 holds with room to spare for the uniform measure.

Remark 24.2 (Why the 2-adic Fourier “noise” was never an obstruction). The $O(1)$ macroscopic peaks of the 2-adic boundary-layer spectrum are the Fourier transform of the *initial Archimedean indicator*, not of the transported measure. The transported conditional measure is exactly Haar; the normalized sum-conditioned 2-adic peaks equal 1 (they do not decay) and *do not need to*: the 2-adic head is exact and cancellation-free.

24.2 The 3-adic head: sum-conditioned superpolynomial decay

Theorem 24.3 (Sum-conditioned Fourier decay (Allikvere, Theorem B)). *Let $\chi(x) = e^{-2\pi i \xi(x \bmod 3^n)/3^n}$, $3 \text{ mid } \xi$. For every $A, C > 0$, uniformly in $|s - 2n| \leq C\sqrt{n \log n}$, $|\mathbb{E}[\chi(\mathbf{X}_n) \mid \mathbf{s}_n = s]| \ll_{A,C} n^{-A}$.*

Observation 24.4 (Adversarial stress test of the 3-adic head; numerical). To close the design gap of the benign first panel, we re-evaluated the sum-conditioned characteristic function of Theorem B (Allikvere) on an adversarial panel consisting of the low-discrete-log family $\xi \equiv 2^k \pmod{3^n}$ — the worst-case resonances of Observation 22.11 — together with random high frequencies, and on the moderate-deviation window $s \in \{2n - \lfloor \sqrt{n} \rfloor, 2n, 2n + \lfloor \sqrt{n} \rfloor\}$, with 10^8 samples per cell. In all nine cells the argmax belongs to the family $\xi = 2^k$ (argmax exponents $k/n \in [1.37, 1.5]$, consistent with the resonance $s^*/n \rightarrow \log_2 3$); the benign panel underestimated the maximum by up to an order of magnitude. The conditional maxima are

n	$s = 2n - \lfloor \sqrt{n} \rfloor$	$s = 2n$	$s = 2n + \lfloor \sqrt{n} \rfloor$
12	0.0734	0.0115	0.0041
16	0.0359	0.0038	0.0008
20	0.0129	0.0020	0.0003

On the central ray the decay over the tested range is $\approx n^{-3}$; the left edge of the window is the noisiest cell, as expected (fewer halvings, weaker equidistribution), yet it also decays: $0.073 \rightarrow 0.036 \rightarrow 0.013$. In contrast to the 2-adic unconditional spectrum, whose peaks stall at $O(1)$ (Observation 18.17), the sum-conditioned 3-adic amplitude decays at every tested scale and every tested deviation, including the resonant family. This is the numerical content of the two-heads dichotomy: the 2-adic head is exact and requires no cancellation; the 3-adic head carries the cancellation, and its worst case — the same low-discrete-log resonance identified independently in Front C — still decays.

Remark 24.5 (Status of the stress test). The stress test is a sanity check of the preprint's Theorem B, not a proof: the superpolynomial rate n^{-A} is an asymptotic upper bound that cannot be confirmed at $n \leq 20$, and the observed central-ray decay $\approx n^{-3}$ over this range is consistent with it. The value of the test is twofold: it identifies the low-discrete-log family as the true bottleneck of the 3-adic head (matching the independent Front C diagnosis), and it shows that even this bottleneck decays, unlike the 2-adic unconditional spectrum.

24.3 Gluing: the natural-density bridge

Corollary 24.6 (Conditional natural-density transport). *Combining the exact 2-adic head with Theorem B and the fibre identity $3^{-n'} \binom{s-1}{n'-1} = 2^t \mathbb{P}(\mathbf{s}_{n'} = s)$, $t = s - n' \log_2 3$, the uniform-measure first-passage law stabilises, $d_{\text{TV}}(\text{Pass}_x(\tilde{\mathbf{N}}_{x^\alpha}), \text{Pass}_x(\tilde{\mathbf{N}}_{x^{\alpha^2}})) \ll \log^{-c} x$, with $c > 1/29$ delivered by the effective irrationality measure $\mu = 8.616$ (Rhin) through the Kronecker-orbit equidistribution; the witnessing iterate occurs within $(3/\ln \frac{4}{3} + 1/\ln 2 + o(1)) \ln N < 12 \ln N$ Collatz steps.*

Remark 24.7 (Dichotomy of cancellations; the threshold $\delta(b) = \log 4 / \log b$). The heads are complementary: the 2-adic head is exact and needs no cancellation; *all* cancellation is carried by the 3-adic head, whose stationary offset measure has full dimension only when $\delta(b) = \log 4 / \log b > 1$, i.e. for $b = 3$. For $b \geq 5$ the 3-adic head is singular, the sum-conditioned decay fails, and the bridge collapses — in exact agreement with the family dichotomy (drift test, dimension collapse, first-passage destruction at $b = 5$).

25 Multi-block survival and unconditional Hausdorff dimension

While the natural-density bridge (Section 8) establishes the stabilization of the first-passage law over $\mathbb{N}$, it does not directly control the Hausdorff dimension of the exceptional set in the 2-adic integers $\mathbb{Z}_2$. The correct metric concept is the 2-adic Hausdorff dimension of the non-descending orbits. We consider the accelerated Syracuse map on blocks of $d = 10$ steps. Let $S(k)$ be the survival probability of an orbit remaining above its initial barrier $N_0 = 2^B$ after k blocks.

Observation 25.1 (Geometric multi-block survival; numerical). We measured the survival curve $S(k)$ for barriers $B \in \{20, 24, 28, 32\}$ and intervals $[2^B, 2^{B\alpha}]$ with $\alpha \in \{1.05, 1.10\}$, simulating 10^7 trajectories up to $T = 1000$ steps per configuration. The survival curve exhibits a clean two-regime structure: an initial transient followed by a stationary geometric decay. The three-parameter fit $S(k) \approx c_*^k \cdot k^{-\gamma}$ cleanly separates the base geometry from the polynomial prefactor, achieving $R^2 > 0.9999$ across all configurations. Crucially, the base survival coefficient $c_*(B)$ does *not* drift to 1 as $B \rightarrow \infty$. Instead, the increments shrink and change sign:

B	$c_*(\alpha = 1.05)$	$c_*(\alpha = 1.10)$	γ
20	0.5250	0.5469	≈ 0.9
24	0.5184	0.5558	≈ 0.8
28	0.5596	0.5605	≈ 0.9
32	0.5385	0.5370	≈ 0.7

Extrapolating $c_*(B)$ against $1/B$ suggests initial stabilization in the narrow band $c_* \in [0.53, 0.58]$ within the observed computational window ($B \leq 32$).¹ The polynomial prefactor $\gamma \approx 0.7 \dots 0.9$ explains the observed growth of the per-block ratio $S(k+1)/S(k) \approx c_*$, resolving the tension between the transient phase and the stationary decay.

¹Note that the extracted base survival $c_* \approx 0.5469$ for $B = 20, \alpha = 1.10$ uses the three-parameter fit isolating the polynomial prefactor γ . A naive two-parameter fit c^k on the same data yields an apparent, smaller rate $c \approx 0.439$, underscoring the necessity of separating the transient from the stationary phase.

25.1 Unconditional block transport and the Hausdorff dimension drop

Setup and conventions (bits; odd classes). Throughout, $\lambda = \log_2 3$, $N_0 = 2^B$, and every rate function is evaluated in bits: $I_2(\sigma) = \sup_s [s\sigma - \Lambda_2(s)]$ with Λ_2 the base-2 cumulant of $\text{Geom}(2)$, so $I_2(1.33) \approx 0.25498$ and $I_2(2) = 0$. (Equivalently $2^{-dI_2(\sigma)} = e^{-dI_{\text{nats}}(\sigma)}$; we never mix the two.) Fix

$$1 < \alpha < \frac{2}{\lambda}, \quad \frac{\alpha - 1}{2 - \lambda} < t \leq \frac{1}{\lambda}, \quad d = \lfloor tB \rfloor, \quad \sigma = \lambda + \frac{\alpha - 1}{t} < 2, \quad m = \lceil \sigma d \rceil,$$

and assume the *parameter gap* $\delta_0 := \alpha - t(\sigma + 1) - tI_2(\sigma) > 0$. The window is nonempty: for $\alpha = 1.10$, $t = 0.30$ one has $\sigma \approx 1.918$, $I_2(\sigma) \approx 0.010$, $\delta_0 \approx 0.22$. Set $c_* := 2^{-BtI_2(\sigma)}$ and write $u_m(r) := 2^{-(m-1)}$ for the uniform (Haar) measure on the 2^{m-1} odd classes modulo 2^m . A block is d accelerated steps; “survival” means all iterates of the block exceed N_0 .

Lemma 25.2 (T1: conditional 2-adic equidistribution of the surviving pushforward). *Let I be a set of M odd integers in an initial window $Y \asymp 2^B$ and let $Q_1(r)$ be the number of $N \in I$ whose first d iterates exceed N_0 and satisfy $\text{Syr}^d(N) \equiv r \pmod{2^m}$. Write $Q_1 = \sum_r Q_1(r)$. Then for every odd $r \pmod{2^m}$, the absolute discrepancy is bounded by the number of actually realized valuation words $\mathcal{W}_{\text{real}}$:*

$$|Q_1(r) - Q_1 2^{-(m-1)}| \leq |\mathcal{W}_{\text{real}}| \leq \min \left(\binom{\sigma d}{d}, \frac{Y^\alpha}{2^{d+1}} + 1 \right).$$

Hence the total variation distance satisfies

$$\|\mu_1 - u_m\|_{\text{TV}} \leq \frac{2^{m-1} |\mathcal{W}_{\text{real}}|}{Q_1} \leq \frac{2^{m-1} \binom{\sigma d}{d}}{Q_1},$$

where $\mu_1(r) = Q_1(r)/Q_1$.

Remark 25.3 (Falsification of the TV convergence via direct summation). To ensure $\|\mu_1 - u_m\|_{\text{TV}} \rightarrow 0$, we require a strict modular constraint $m + \log_2 |\mathcal{W}_{\text{real}}| < \log_2 Q_1$, which translates to a parameter gap requirement $\delta_1 := \alpha - \sigma t(1 + H(1/\sigma)) + tI_2(\sigma) > 0$. However, substituting the operational values $\sigma \approx 1.918$, $t = 0.3$, $\alpha = 1.10$, and $I_2(1.918) \approx 0.010$ yields $\delta_1 \approx -0.052 < 0$. Thus, the TV bound provided by Lemma 25.2 is *vacuous* for large B . Direct summation of intra-word discrepancies is too loose to establish the TV proximity required for restart. The restart/transport hypothesis remains deeply open, demanding a non-discrepancy mechanism.

Proof. By the exact affine word identity $\text{Syr}^d(N) = y_w + 2 \cdot 3^d q$ on the cylinder of a word w (Lemma 18.12), the barrier conditions cut q to an integer interval J_w , and since 3^d is a unit modulo 2^{m-1} , the map $q \mapsto y_w + 2 \cdot 3^d q \pmod{2^m}$ runs through the odd classes with period 2^{m-1} . Counting an interval in a fixed odd class gives discrepancy at most one: $\#\{q \in J_w : y_w + 2 \cdot 3^d q \equiv r\} = |J_w| 2^{-(m-1)} + O(1)$. Summing the per-word discrepancy ≤ 1 over the realized words gives $|Q_1(r) - Q_1 2^{-(m-1)}| \leq |\mathcal{W}|$, where $|\mathcal{W}| \leq \binom{\sigma d}{d}$ by the hockey-stick identity $\sum_{S=d}^m \binom{S-1}{d-1} = \binom{m}{d}$ (formalized in Lean as `W1_word_count_bound`). At the working parameters this bound is *vacuous* ($\delta_1 < 0$), consistent with the falsification of sum-conditioned TV convergence (Remark below). Dividing by Q_1 and summing over the 2^{m-1} odd classes gives the TV bound. (The comparison is against the odd classes only.) $\square$

Lemma 25.4 (T2: uniform one-block descent and Lyapunov drift). *Let μ be a probability measure on odd integers with $\text{supp } \mu \subset [N_0, N_0^U]$ and $\|\mu - u_m\|_{\text{TV}} \leq \varepsilon$. Let $P_{\text{surv}}(\mu)$ be the μ -mass of survival over one block and $x^+ := \text{Syr}^d(x)$ on survival. Then:*

(i) $P_{\text{surv}}(\mu) \leq c_* + 2\varepsilon$;

(ii) for $V(x) := (x/N_0)^\beta$ with $0 < \beta \leq \beta_0(B, U)$ there exist $u_0, w > 0$ and $\delta > 0$ such that

$$\int_{\text{surv}} V(x^+) d\mu(x) \leq \rho \int V d\mu + C 2^{-\delta B}, \quad \rho := \max((c_* + 2\varepsilon)2^{\beta u_0 B} + 2^{\beta w B}, 2^{-\delta}) < 1.$$

Proof. (i) If all d iterates exceed N_0 , the exact identity $x^+ = x3^d 2^{-S_d} \prod_{k < d} (1 + \frac{1}{3x_k})$ forces $S_d < \sigma d$ (as in Corollary 18.11). The exact word cylinders $\{\text{cyl}_w : |w| \leq \sigma d\}$ are disjoint residue classes, so $P_\mu(S_d \leq \sigma d) = \sum \mu(\text{cyl}_w) \leq \sum 2^{-S} + 2\varepsilon = 2^{-dI_2(\sigma) + O(\log d)} + 2\varepsilon = c_* + 2\varepsilon$, using Proposition 18.10 with I_2 in bits. (ii) Write $x = N_0^v$, $v \in [1, U]$. On survival, $x^+ = x3^d 2^{-S_d} (1 + O(N_0^{-1}))$. Split by the event $\mathcal{R}_w := \{S_d \geq d\lambda + \log_2(x/N_0) - wB\}$: on $\mathcal{R}_w$, $x^+ \leq N_0^w (1 + o(1))$, hence $V(x^+) \leq 2^{\beta w B}$. On the complement, $S_d \leq d\lambda + (v - w)B$, and Proposition 18.10 (at $m_u = d\lambda + (v - w)B \leq \sigma d$ for v in the support) gives μ -probability $\leq 2^{-dI_2(\sigma_w)}$ with $\sigma_w = \lambda + (v - w)/t$; there $V(x^+) \leq V(x)3^{\beta d}$. For $v \geq u_0$ with $u_0 > w$ fixed, $I_2(\sigma_{u_0}) > 0$ uniformly, so choosing β small enough makes $3^{\beta d} 2^{-dI_2(\sigma_{u_0})} \leq 2^{-\delta B}$ and the rare-set contribution is $\leq 2^{-\delta B} \int V d\mu$. For $v < u_0$, $V \leq 2^{\beta u_0 B}$ and the mass is $\leq P_{\text{surv}} \leq c_* + 2\varepsilon$ by (i). Collecting, $\int_{\text{surv}} V(x^+) \leq (c_* + 2\varepsilon)2^{\beta u_0 B} + 2^{\beta w B} + 2^{-\delta B} \int V d\mu$. Since $c_* < 1$, choosing β so that $(c_* + 2\varepsilon)2^{\beta u_0 B} + 2^{\beta w B} < 1$ gives $\rho < 1$. $\square$

Remark 25.5 (TV-free variant of (ii)). The hypothesis $\|\mu - u_m\|_{\text{TV}} \leq \varepsilon$ in (ii) is used only through (i). Replacing $c_* + 2\varepsilon$ by the TV-free bound $c_* + \binom{\sigma d}{d}/(A_k M)$ from Proposition 18.10 makes (ii) unconditional; Theorem 25.6 uses this TV-free form and does not depend on the vacuous TV bound of Lemma 25.2.

Theorem 25.6 (T3: geometric multi-block survival; unconditional $\dim_H < 1$). *Let $E_{N_0} = \{N \in 2\mathbb{N} + 1 : \text{Syr}^j(N) > N_0 \ \forall j \geq 1\}$. For any 2-adic ball $\mathcal{B}$ of radius $2^{-\alpha B}$ (i.e. $M = 2^{\alpha B - 1}$ odd starts), the mass A_k of $E_{N_0} \cap \mathcal{B}$ surviving k blocks satisfies the TV-free recurrence $A_{k+1} \leq c_* A_k + \binom{\sigma d}{d}/M$, hence $A_k \leq C c_*^k + C' 2^{-\delta_d B}$ with $c_* < 1$ and $\delta_d = \alpha - \sigma t H_2(1/\sigma) > 0$. Consequently*

$$\dim_H(E_{N_0}) \leq 1 + \frac{\log_2 c_*}{2d} < 1,$$

and E_{N_0} has 2-adic Haar measure $\leq C' 2^{-\delta_d B}$, exponentially small in B ; in particular the divergent set $\bigcap_{N_0} E_{N_0}$ has 2-adic Haar measure zero.

Proof. Let A_k be the total mass of survivors after k blocks in $\mathcal{B}$, so the number of surviving odd classes is $|I_k| = A_k M$ with $M = 2^{\alpha B - 1}$. Applying the finite-scale 2-adic large-deviation principle (Proposition 18.10) directly to the set of survivors I_k with $m = \lceil \sigma d \rceil$ yields:

$$\frac{|I_{k+1}|}{|I_k|} \leq 2^{-dI_2(\sigma) + O(\log d)} + \frac{\binom{\sigma d}{d}}{|I_k|} = c_* + \frac{\binom{\sigma d}{d}}{A_k M}.$$

Multiplying by A_k gives the recurrence:

$$A_{k+1} \leq c_* A_k + \frac{\binom{\sigma d}{d}}{M}.$$

This additive error bypasses the vacuous TV-error $2^{m+d+1}/M$ (which failed since $\delta_1 < 0$) with the much smaller word-counting bound $\binom{\sigma d}{d}/M \leq 2^{1-\delta_d B}$, where the parameter gap $\delta_d = \alpha - \sigma t H_2(1/\sigma) \approx 0.5255 > 0$. Iterating the TV-free recurrence $A_{k+1} \leq c_* A_k + \binom{\sigma d}{d}/M$ with $A_0 \leq 1$ gives

$$A_k \leq c_*^k + \frac{\binom{\sigma d}{d}}{M} \cdot \frac{1 - c_*^k}{1 - c_*} \leq c_*^k + C' 2^{-\delta_d B}, \quad C' = \frac{2}{1 - c_*},$$

using $\binom{\sigma d}{d}/M \leq 2^{1-\delta_d B}$. Note that the residual is a **constant** in k , not linear: the geometric series is summable, so the floor freezes at height $C' 2^{-\delta_d B}$ rather than growing with k . (The Lyapunov route of Lemma 25.4(ii) yields the same form with its own ρ, δ ; the present argument uses the TV-free form and does not depend on the vacuous TV bound of Lemma 25.2.) This frozen residual $C' 2^{-\delta_d B}$ is the *resolution floor* of scale B : it is the mass that the contraction cannot squeeze out within a single block-resolution of the barrier. Exponential decay $C c_*^k$ is effective only while $C c_*^k \gg C' 2^{-\delta_d B}$, i.e. for $k \ll \delta_d B / |\log_2 c_*|$. Beyond this depth the bound is dominated by the resolution floor, and the s -content cannot be driven to zero by sending $k \rightarrow \infty$ at fixed B . Consequently the Hausdorff dimension bound is established by choosing $k = \Theta(B)$ so that $c_*^k \asymp 2^{-\delta_d B}$, rather than by taking $k \rightarrow \infty$. For the dimension, cover $E_{N_0} \cap \mathcal{B}$ at depth k by the surviving word cylinders, each of modulus 2^{S_k} with $S_k \geq dk$ and Haar diameter 2^{-S_k} . Their s -content is $\leq A_k 2^{(1-s)dk} \leq (C c_*^k + C' 2^{-\delta_d B}) 2^{(1-s)dk}$. Choosing $s = 1 + \frac{|\log_2 c_*|}{2d} - o(1)$ and $k = \Theta(B)$ with $c_*^k \asymp 2^{-\delta_d B}$ makes the s -content $\leq 2C' 2^{-\delta_d B} c_*^{k/2} \asymp 2^{-3\delta_d B/2} \rightarrow 0$ as $B \rightarrow \infty$. Thus $\dim_H(E_{N_0}) \leq 1 + \frac{\log_2 c_*}{2d} < 1$ for each fixed N_0 . The bound is for fixed N_0 ; it does not yield a uniform dimension bound independent of the barrier, and it does not imply that E_{N_0} is empty. The bound is uniform over balls $\mathcal{B}$ of radius $2^{-\alpha B}$, and $\mathbb{Z}_2$ is covered by finitely many such balls per scale; letting the scale tend to the fixed N_0 -resolution gives $\dim_H(E_{N_0}) \leq 1 + \log_2 c_* / (2d) < 1$. Letting $k \rightarrow \infty$ in $A_k \leq C c_*^k + C' 2^{-\delta_d B}$ gives $\mu(E_{N_0} \cap \mathcal{B}) \leq C' 2^{-\delta_d B}$, i.e. E_{N_0} has 2-adic Haar measure $\leq C' 2^{-\delta_d B}$, exponentially small in B but not asserted to be zero for fixed B . Since this holds for every B , the divergent set $\bigcap_{N_0} E_{N_0}$ has 2-adic Haar measure zero. $\square$

Remark 25.7 (Rules honoured; scope). All TV comparisons are against $u_m(r) = 2^{-(m-1)}$ on the odd classes. All rates $c_* = 2^{-BtI_2(\sigma)}$ are with I_2 in bits. The value $c_* = 2^{-BtI_2(\sigma)}$ is an *upper bound* on the one-block survival probability; the measured stationary band $c_* \in 0.51\text{--}0.56$ (Observation above) lies below it, consistent with the recurrence. The numerical band is a finite-scale observation and is not used as a premise of any theorem. Theorem 25.6 is unconditional for each *fixed* N_0 . It does not imply the full pointwise conjecture: a counterexample lies in $\bigcap_{N_0} E_{N_0}$, a countable intersection of $\dim_H < 1$ sets, which may still be non-empty.

25.2 Natural-density polylogarithmic descent

The deterministic 2-adic transport assembly (first-passage reversal over decreasing thresholds) was recently shown by Shaik [8] (August 2026) to unconditionally bound the measure of exceptional trajectories in natural density. Our exact conditional transport (Theorem 23.2) natively acts as the strict topological input for that assembly.

Corollary 25.8 (Unconditional polylogarithmic descent in natural density). *For almost all N in natural density, the minimum of the Collatz orbit satisfies*

$$\text{Col}_{\min}(N) \leq C(\log N)^A$$

for any $A > A_{FP} \approx 9.991$, achieved within $c \log N$ unaccelerated steps (for $c > 3/\log(4/3) \approx 10.428$), with the entire orbit path uniformly bounded by $N^{1+\beta}$ for any $\beta > 0$.

Proof. As demonstrated in recent work [8], assembling the 2-adic reverse transport along a chain of geometrically decreasing thresholds reduces the union bound of passage times from a linear horizon $O(M)$ to a square-root corridor $O(\sqrt{M \log M})$. Combined with the subcritical binomial timeout tail $\kappa_* = 1 - H_2(\log_3 2) \approx 0.05$ (the exact rate of which we independently verified numerically), this yields the critical exponent $A_{FP} = (2\kappa_*)^{-1}$. This route is fully deterministic and independent of the sum-conditioned 3-adic Fourier decay (Theorem B), closing the almost-all gap unconditionally. $\square$

Remark 25.9 (Honest status and synthesis). The point-wise Collatz conjecture remains open. The contribution of this programme is fivefold: the exact 2-adic transport theorem, the numerical validation of the 3-adic pillar, the synthesis exhibiting why the two heads must be treated by different methods, the unconditional Hausdorff dimension drop for the exceptional set at any fixed barrier, and the formal bridging to Shaik’s [8] unconditional natural-density polylogarithmic descent. While 2-adic transport alone determines the bounding exponents, the 3-adic Fourier spectrum (Allikvere’s Theorem B [9]) and our adversarial stress test remain structurally necessary for establishing the *exact distribution* of the first-passage law.

26 Research and Software Disclosure

Generative AI systems (specifically Anthropic Claude/Claude Code, DeepSeek, Alibaba Qwen, and Google Gemini) were used throughout this project for the following tasks:

- **Computational exploration:** Development and debugging of Python scripts for parity scanning (`zone_parity_search.py`), confluence census (`confluence_census.py`), reverse tree generation (`reverse_tree_xstar.py`), Fourier spectral analysis, the Gate 2 transport-mechanism campaign (`restart_discrepancy_decay.py`, `timeout_renewal_closure.py`, `wasserstein_multiblock.py`), and the cycle-exclusion experiments (`cycle_fixedpoint_census.py`, `cycle_highmean_admissibility.py`, `cycle_baker_exclude.py`).
- **Formal verification:** Lean 4 formalization of structural theorems (Proposition B3, Exact Conditional Transport, multi-block survival recurrence), including syntax generation, proof skeleton construction, and type-checking workflows.
- **Diagnostic analysis:** Numerical stress testing of Fourier decay hypotheses (Observations 18.17, 22.11), polymer free energy modeling (Observation 22.13), adversarial validation of sum-conditioned cancellation, and the Gate 2 falsification diagnostics.
- **Structural synthesis:** Organization of the five-result framework (F1/R1, S1, I1, C1, M1), identification of closed proof paths (Low-Frequency Cancellation, Spectral Transversality, Reverse-chord encoding, and the three Gate 2 mechanisms), and formulation of the two-heads dichotomy (Section 24).

The authors selected all mathematical architectures, verified all theoretical claims, performed independent validation of numerical results, and accept full responsibility for the content of this paper. No finite computation serves as a premise of any theorem; all numerical evidence is explicitly labeled as observation or diagnostic. All code, data, and Lean formalizations are available in the supplementary materials and at the project repository.

27 Conclusion: What Was Mapped, What Was Closed, and What Remains

This programme set out to understand the Collatz space through the lens of large-deviation theory, and it ends not with a proof of the conjecture but with something we believe is equally valuable: a *complete map of the territory*, with every false path explicitly closed and every open bridge precisely identified.

The five linked results. The theoretical core of the paper is the duality between the *typical* sector (decaying orbits) and the *rigid* sector (confluence structures), governed by a single rate function. Concretely: the forward shift-vector obeys a Cramér LDP with the geometric(2) rate I_2 (**F1**); the tilted 3-adic reverse operator satisfies a backward LDP whose normalized rate coincides with I_2 (**R1**); the generic peak-path is a *soliton* that avoids all rigid structures (**S1**); every confluence center is a *dressed vacuum*—the $(1, 1, 2)$ fixed point $\xi = -29/11$ with $O(1)$ total charge (**I1**); each center sits at the log-midpoint $\text{bits}(c) = \frac{1}{2}P + O(1)$ by detailed balance at the waist (**C1**); and constructive reverse synthesis of macro-centers is algorithmically closed by a CRT dimensionality deficit $\Delta \geq 2P$ (**M1**). The paradigm crystallises as: *crystals can be found but not grown*.

The closure of the Fourier route. The most consequential negative result of this programme is the systematic falsification of every candidate Fourier mechanism for the restart/transport step. The Low-Frequency Cancellation Lemma (macroscopic peaks ~ 0.6 at $M = 22, 24$ with no decay), Spectral Transversality (8/10 unrestricted peaks aligning with the dominant layer), and the sum-conditioned restart hypothesis (normalized peaks = 1.0000, the apparent $O(d^{-1/2})$ decay exposed as a Gaussian mass artifact) are all conclusively closed (Observations 18.17, 18.18; Section 24). The two-regime structure of the 3-adic Fourier spectrum (fast bulk L^2 decay $c_{\text{avg}} \approx 0.61$ versus worst-case $c_\infty \leq 0.054$ with $c_\infty = 0$ not excluded; Observation 22.11) and the worst-case barrier (Theorem 22.14, cf. Tao Remark 1.18) establish that no L^2 averaging can bypass the pointwise Fourier rate. The polymer free-energy diagnostic (Observation 22.13) provides numerical evidence that $c_\infty > 0$ but with a microscopic rate $\sim 10^{-6}$, confirming that the 3-adic head is theoretically sound yet practically too slow to bridge the Archimedean gap alone.

The Gate 2 campaign: closure of the pointwise transport route. Beyond the Fourier route, the Gate 2 campaign (Section 19) systematically falsifies the three surviving candidate pointwise transport mechanisms: TV–Fourier restart (G2.1), renewal closure (G2.2), and Tree–Wasserstein multiblock contraction (G2.3). The verdict is unambiguous: the Archimedean–2-adic wall stands, and the pointwise martingale/transport route is closed. Every mechanism proposed for bridging the local 2-adic descent across scales is now either falsified or shown to be mechanism-free.

Cycle exclusion. Three independent arguments converge on the exclusion of non-trivial cycles up to length 10^6 : a fixed-point census (0 non-trivial solutions at every reachable length), a 3-adic dimension-collapse argument (cyclic shift-words occupy a measure-zero set, $D_1 \approx 0.76$), and a Baker-type Diophantine argument combined with

the Barina frontier 2^{68} (Section 20). These results do not rule out cycles of length $> 10^6$, but they establish that any non-trivial cycle must be extraordinarily long.

The architectural contribution. The central structural insight is the *separation of concerns between two independent heads*. The 2-adic head is *exact and cancellation-free*: the affine word map T_w bijects each cylinder onto $\mathbb{Z}_2$ (Theorem 23.2), the conditional endpoint law on every S -layer is exactly Haar, and the $O(1)$ Fourier “noise” of the unrestricted spectrum is entirely the Fourier transform of the initial Archimedean indicator, not a mixing failure. The 3-adic head carries *all* the cancellation: its sum-conditioned Fourier decay (Allikvere Theorem B) is superpolynomial and survives the adversarial stress test (Observation 24.4), but it is specific to $b = 3$ and fails for $b \geq 5$ where the stationary IFS measure is singular. The threshold $\delta(b) = \log 4 / \log b \gtrsim 1$ is the exact condition for the Allikvere transport to operate. This two-heads architecture explains why the natural-density bridge can stand for $3x + 1$ and must collapse for $5x + 1$.

New unconditional results. Three unconditional results emerge from the synthesis. *First*, the multi-block survival analysis (Section 24, Theorem 25.6) establishes $\dim_H(E_{N_0}) \leq 1 + \log_2 c_*/(2d) < 1$ for each fixed barrier $N_0 = 2^B$, with the corrected iteration showing that the residual is a *constant* resolution floor $C'2^{-\delta_d B}$, not a growing term; the Hausdorff bound is taken at the correct depth $k = \Theta(B)$. *Second*, the exact 2-adic transport (Lemma 25.2) provides the strict input for Shaik’s unconditional natural-density assembly, yielding $\text{Col}_{\min}(N) \leq C(\log N)^A$ for $A > A_{\text{FP}} \approx 9.991$ in natural density (Corollary in Section 24). *Third*, the family dichotomy $\delta(b) = \log 4 / \log b$ is validated by three independent numerical tests (drift, dimension collapse, first-passage destruction) and identifies $b = 3$ as the unique value where the transport mechanism operates.

What remains open. Four bridges remain open, and we state them precisely. *(i)* The forward restart/transport hypothesis across scales is open and mechanism-free: every candidate mechanism — pointwise Fourier decay, low-frequency/aggregate cancellation, spectral transversality, TV–Fourier restart, renewal closure, and Tree–Wasserstein multi-block contraction — has been falsified (Gate 2). The Archimedean–2-adic misalignment is the true obstruction that logarithmic density bypasses and natural density must attack via Kronecker orbits and transcendence theory. *(ii)* The worst-case Fourier rate c_∞ is not determined; if $c_\infty = 0$, Tao’s architecture yields a superpolynomial but not an exponential exceptional-set bound, and the effective threshold N_0^* remains astronomically above the computational frontier 2^{68} . *(iii)* The pointwise Collatz conjecture $\text{Col}_{\min}(N) = 1 \forall N$ is not implied by any density bound: a counterexample lies in $\bigcap_{N_0} E_{N_0}$, a countable intersection of $\dim_H < 1$ sets that may still be non-empty. *(iv)* The reverse-chord encoding is superseded by an intrinsic entropy deficit ($h(4/3) \approx 1.0817 < \log_2 3 \approx 1.585$), closing the route to attaching $\gamma_{\text{rev}} \approx 0.993$ to the forward exceptional set.

The path forward. The map drawn here suggests three directions. *Transcendence/Kronecker*: the only open bridge to a pointwise result is the Archimedean–2-adic mismatch attacked via Kronecker-orbit equidistribution and transcendence theory; a counterexample, if it exists, must lie in this mismatch. *The c_∞ question*: closing $c_\infty > 0$ rigorously (via the polymer free-energy or a renewal-theoretic argument) would upgrade the superpolynomial bound to exponential, though the microscopic rate suggests the effective gain would

be modest. *Cycle exclusion at larger lengths*: extending the Baker-type exclusion beyond 10^6 , combined with the fixed-point census and dimension collapse, would tighten the cycle side of the conjecture.

Final word. We have not proved the Collatz conjecture. What we have done is to distinguish, for the first time in a single coherent framework, the five different *types* of statement that the word “Collatz” conflates: the typical decay (proven by Tao), the natural-density descent (proven in natural density by Shaik [8]; the Tao-to-natural-density bridges of Allikvere and Mazur [9, 10] remain unpublished), the Hausdorff dimension of the exceptional set (established here for fixed barriers), the structural anatomy of the rare rigid objects (mapped here in detail), and the pointwise conjecture itself (open since 1937). The first four are now cleanly separated; the fifth is seen clearly across the gap. Whether that gap can be crossed is a question for mathematics that does not yet exist.

28 Formal Machine Verification and Diophantine Obstructions

To place our findings on an absolutely rigorous foundation, the algebraic backbone of this paper—including the fractional bounds on macroscopic cycles and the large-deviation topology of divergent orbits—has been formalized in the Lean 4 interactive theorem prover (see the `CollatzLean` repository). This formalization achieves zero `sorry` gaps in its core algebraic layers and cleanly delineates the boundaries of cycle exclusion versus the pointwise divergence trap.

28.1 Algebraic Bedrock in Lean 4

The foundational arithmetic of the Collatz 2-adic transport has been fully verified. We have formalized the shift-vector structure, the CRT dimensionality obstruction (Theorem M1), and the localized descent mechanisms (the “Shadow Mountain” descent bounds). The machine verification guarantees that our algebraic setup—including the precise fractional bounding $0 < S \ln 2 - K \ln 3 < K/x_{\min}$ for cycles—contains no hidden continuous-variable approximations or modular arithmetic artifacts.

28.2 Strict Cycle Exclusion: Continued Fractions vs. Transcendental Bounds

Our framework formally distinguishes between two distinct cycle-exclusion mechanisms, avoiding the category error of mixing probabilistic dimension-collapse ($D_1 \approx 0.76$) with rigid Diophantine bounds:

1. The Computational Frontier (Continued Fractions). For practical cycle exclusion (lengths $d \leq 10^{11}$), transcendental asymptotics are excessively weak. Instead, our framework leverages the continued fraction expansion of $\ln 3 / \ln 2$ (following Eliahou, 1993). Because the convergents p/q provide the tightest possible approximations $|q \ln 2 - p \ln 3| \approx 1/q$, any macroscopic cycle length d must align with a convergent denominator. Combined with Barina’s verified frontier $x_{\min} > 2^{68}$, the continued fraction method rigorously excludes all non-trivial cycles of length $d \leq 1.14 \times 10^{11}$.

2. The Asymptotic Frontier (Baker–Rhin). For the limit $d \rightarrow \infty$, we formalize the Baker–Rhin bound for linear forms in logarithms (packaged in our `BakerRhin.lean` interface). The bound $\mu = 8.616$ enforces that the spacing $|S \ln 2 - d \ln 3| \geq C \cdot d^{-7.616}$ cannot shrink fast enough to satisfy the cycle requirement $1/x_{\min}$, proving that macroscopic cycles are rigorously asymptotically dead.

28.3 The Combinatorial Formalization of Terras (1976)

As a capstone to the probabilistic mapping, we have formalized Terras’ 1976 Theorem (almost all integers have a finite stopping time) in Lean 4 (`Terras.lean`). Crucially, to bypass the lack of continuous stochastic PDEs in Mathlib, our proof relies entirely on the discrete combinatorics of the boolean cube and affine shift properties. This establishes the structural density drop without invoking continuous measure-theoretic approximations, proving that the proportion of “bad vectors” collapses strictly.

28.4 The Epistemological Wall: Why Measure Zero $\neq \emptyset$

The machine verification explicitly documents the Conway (1972) undecidability barrier. It is mathematically critical to distinguish between cycles and divergent orbits in the context of Diophantine rigidity:

- **Cycles:** A cycle requires the multiplier $3^d/2^S$ to approach 1 with phenomenal precision (to compensate for the massive $x_{\min}$). Baker’s Theorem and continued fractions explicitly forbid this hyper-accurate convergence. Cycles are killed by Diophantine rigidity.
- **Divergent Orbits:** A divergent orbit (an infinite “needle”) only requires $3^d > 2^S$ (with a macroscopic margin). Divergent orbits do not require Diophantine convergence, meaning Baker’s Theorem is entirely powerless against them.

While Terras’ theorem guarantees that the surviving set of divergent genealogies forms a fractal of Haar measure 0 in $\mathbb{Z}_2$, the natural numbers $\mathbb{N}$ are dense in $\mathbb{Z}_2$. Any finite-prefix approach (including our CRT dimensionality mapping) requires an infinite sequence of exact modular alignments to track a specific natural number $N \in \mathbb{N}$. Testing whether a fixed integer threads the infinite needle requires an unbounded state space, running headfirst into the Measure-Zero Trap (Conway’s Undecidability). Our framework does not claim to bypass this wall; rather, it rigorously defines its exact mathematical coordinates.

Data Availability and Formalization

The full Zone 2 catalog (913 entries, `zone2_shifts_full.csv`), 39-center census, KL derivation, beam-search death logs, Gate 2 transport-mechanism logs, cycle-exclusion logs, algorithmic pseudocodes, and Python verification scripts are available in the project GitHub repository (<https://github.com/SergioTheory/Collatz-new-math>). The Lean 4 formalization of the exact conditional transport (Theorem 23.2) and the CRT dimensionality obstruction (Theorem M1) are also provided in the repository.

References

- [1] Lagarias, J.C. (1985). *The $3x+1$ problem and its generalizations*. Amer. Math. Monthly 92(1), 3–23.
- [2] Tao, T. (2022). *Almost all orbits of the Collatz map attain almost bounded values*. Forum Math. Sigma 10, e12. (arXiv:1909.03562.)
- [3] Barina, D. (2021). *Convergence verification of the Collatz problem*. J. Supercomputing 77, 2681–2688.
- [4] Dembo, A., Zeitouni, O. (1998). *Large Deviations Techniques and Applications* (2nd ed.). Springer.
- [5] Kontorovich, A., Miller, S.J. (2005). Benford’s law, values of L -functions and the $3x+1$ problem. *Acta Arith.* 120, 269–297.
- [6] Korec, I. (1994). A density estimate for the $3x+1$ problem. *Math. Slovaca* 44(1), 85–89.
- [7] Terras, R. (1976). A stopping time problem on the positive integers. *Acta Arith.* 30, 241–252.
- [8] Shaik, I.A. (2026). *Polylogarithmic Descent for Almost All Collatz Orbits in Natural Density*. Preprint v3.2.3. DOI: 10.5281/zenodo.21984038.
- [9] Allikvere, J. (2026). Natural-density bridges from Tao’s logarithmic-density framework. Unpublished manuscript.
- [10] Mazur, L. (2026). Quantitative phase-gap and 2-adic lift from the logarithmic-density frontier. Unpublished manuscript.

A Main Project Scripts

Core: `crt_solver.py`, `records_data.py`. **Search:** `peak_hunter.py`, `zone_parity_search.py`, `residue_search.py`, `class_a_search_51_80.py`. **Zone 2:** `confluence_search.py`, `reverse_tree_xstar.py`, `count_zone2_in_tree.py`, `analyze_barina.py`. **Centers:** `confluence_census.py`, `verify_census_centers.py`, `targeted_search_31_50.py`, `targeted_search_41_50.py`, `algebra_centers.py`, `two_class_analysis.py`, `basin_test.py`. **Spectrum:** `family_a_spectrum.py`, `check_microplateau.py`, `analyze_records_gain.py`, `analyze_27.py`. **Automata:** `automaton_invariants.py`, `automaton_compare.py`. **Gate 2:** `restart_discrepancy_decay.py`, `timeout_renewal_closure.py`, `wasserstein_multiblock.py`. **Cycle exclusion:** `cycle_fixedpoint_census.py`, `cycle_highmean_admissibility.py`, `cycle_baker_exclude.py`.

B Key Algorithmic Primitives

1. **Confluence filter:** $c \equiv 1 \pmod{2}$, $c \equiv 2 \pmod{3}$, $v_2(3c + 1) = 1$, $|\text{bits}(c) - (0.498P + 6.29)| \leq 3$ (rejects $\approx 5/6$).

2. **Reverse step:** $T^{-1}(x, a) = \{(x2^a - 1)/3\}$, $a = 1..15$, odd positive only.
3. **Hit rate:** $HR(c, P; d_{win}, W) = |\{y \in V_{d_{win}} \cap W : \text{peak}(y) = P\}|/|V_{d_{win}} \cap W|$.
4. **Class A:** $HR = 1.0$ for some window, $d_{peak} > 20$, $S/d_{peak} > 1.30$.
5. **Gate 2 signed/absolute ratio:** $|\sum_w \epsilon_w|/\sum_w |\epsilon_w|$ (cancellation diagnostic; 0.5–0.9 observed, far from cancellation).
6. **Cycle fixed-point test:** solve $(2^S - 3^d)x = c_d$ for positive integer x over admissible words $\vec{a}$ of length d .

C Nature of the Instanton (Expedition B)

Lemma C.1 (2-adic vacuum). $I = I_2 \circ I_1 \circ I_1$ is a contraction on $\mathbb{Z}_2$ (ratio 2^{-4}) with unique fixed point $\xi = -29/11$, generating a periodic inverse orbit with $S/d = 4/3$. Finite orbits shadowing ξ to 2^{-m} over d steps have $S/d = 4/3 + O(1/d) + O(2^{-m})$.

Lemma C.2 (Gain/charge balance). For w of length d , $S = \frac{4}{3}d + \delta$: $G(d) = d(\log_2 3 - \frac{4}{3}) - \delta$. Zone 2 core ($d = 251$, $P - B \approx 68.4$): $\delta \approx -0.67$ (low defect density).

Hypothesis B3 (center as log-midpoint). For Class A centers, $\text{bits}(c) \approx P/2 + 6.5$; the center is the deepest common node with sufficient cylinder measure. (Rigorous form: Part II, Theorem C1.)

D Lemma Ledger

The following ledger records the status of every major computational and theoretical claim.

Date	Script Name	Status	Key Number	Interpretation
2026-08-24	1.1 cycle_fixedpoint_census.py	numerical	0	0 non-trivial solutions
2026-08-24	1.2 cycle_highmean_admissibility.py	numerical	$D_1 < 1$	Dimension collapse
2026-08-24	2.1 restart_discrepancy_decay.py	numerical	$p < 0$	[GATE-2] TV-Four
2026-08-24	2.2 timeout_renewal_closure.py	numerical	c_* unstable	[GATE-2] Renewal
2026-08-24	2.3 wasserstein_multiblock.py	numerical	$\rho > 1$	[GATE-2] W1 mult
2026-08-24	VERDICT	final	—	[GATE-2] VERDICT
2026-08-24	3.1 cycle_baker_exclude.py	analytic	$> 10^6$	All cycles of length